

Memory, Recurrent Geometry, and Rendezvous Traps in Induced Grid Mazes

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Abstract

Two identical deterministic finite-state agents move synchronously and compulsorily in a finite connected induced subgraph of the square grid. They share an absolute compass, observe the open-neighbour mask, and start in one distinguished state. Rendezvous means attaining Manhattan distance at most one at an integer time. We relate the memory budget to recurrent geometry, global transition compatibility, and the size and uniformity of rendezvous obstructions. A support-capacity law bounds the multiplicities of a least-period tagged walk on a tree, and an explicit family attains the sharp maximum path period for every state budget. The exact smallest-trap thresholds are $F(1) = 4$ and $F(2) = 7$, with the two-state value unchanged on induced grid trees; both are proved without computational assistance. On induced paths the two-state threshold is eight, with an analytic upper bound and an exact finite lower certificate, whereas one fixed four-state controller succeeds on every path. For three states we give an all-size constructive realization criterion for prescribed contour-plus-matching cycles: a finite local profile selects a sparse affine system over $\mathbb{F}_2$, whose consistency is necessary and sufficient. Non-affine phase projections show why one global XOR system is insufficient. Finally, the two-state obstruction proof yields an explicit finite complete test family and, through protected attachment, one universal induced subcubic grid-tree host. Recurrent-cycle realization is distinguished from initial-state accessibility; universal three-state rendezvous remains open.

1 Introduction

Finite-state navigation connects automata in labyrinths [2, 19], graph exploration [9, 7] and deterministic rendezvous [22, 11, 4]. Their common memory constraint governs both the geometry of eventual motion and the compatibility of local decisions. Distinct visits to a cell on a least-period recurrent orbit need distinct states; on a tree this bounds incident edge multiplicities. Spatially distant cells with the same compass mask impose a second, global constraint: their visits must use one common controller table.

We study identical agents moving synchronously in finite connected induced grid mazes. Each observes the open compass directions and must choose one; both start in a distinguished state. Success means coincidence or adjacency at an integer time. There is no waiting, marking, communication, arrival-port observation or continuous-time meeting event. Exploration failure need not keep two moving copies apart, and a recurrent cycle need not be accessible from the initial state. These distinctions also govern our comparison with finite-automaton rendezvous results [5].

For each controller, take its smallest trap and then maximize over controllers with at most s states: this defines $F(s)$. The adversary chooses the maze and starts after the controller is fixed. We also treat restricted maze classes, arbitrary success radii, and the dual memory parameter $M_{C,\rho}(n)$ for solving every instance through size n .

Memory, geometry and compatibility. Degree constraints and contour traversal have exploration predecessors [6, 17]; our structural object is a weighted recurrent support. Its capacity law yields a sharp maximum path period for every $s \geq 2$: $sn - 2$ for even $n \geq 3$, and $s(n - 1)$ for odd $n \geq 3$. Attainment is over geometries and controllers, not on every prescribed path. Two phases of one

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recurrent path cycle must attain distance one; branching trees need not have this property. For three states, capacity permits a contour with doubled edges forming a matching. Realization by one table requires more. In the setting neighboring local state annotations and scenario synthesis [21, 23], we reduce it to one shared finite profile and a sparse affine system over $\mathbb{F}_2$. Eliminating the profile can produce NAND, so a single XOR system is insufficient. The criterion concerns a cycle on the path with its own masks, not its basin.

Sharp low-state thresholds. Earlier exact low-state exploration results use port-based models [10]. For our distance-one problem, $F(1) = 4$ belongs to the radius law $\lfloor \rho \rfloor + 3$ for $\rho \geq 1$ on paths, one-junction trees, grid trees and general grid mazes; the coincidence threshold is two. The technical centre is a proof of $F(2) = 7$ without computational assistance, unchanged on induced grid trees. Small path gadgets force a nine-entry pattern, and two junction completions give the upper bound. An explicit controller A^* supplies the reverse bound through recurrent-support classification, initial-state gates and transient contact. On paths the threshold is eight, with an analytic upper bound and an exhaustive finite lower certificate. Four states suffice universally on paths; the three-state question remains open.

From a test menu to one host. Classical universal traps already strengthen individual-obstruction quantifiers [1, 18]. Here one must preserve distance-one failure and every queried compass mask. The two-state case proof yields an explicit complete test family; a certified smaller selection is optimal only within those candidates. Protected attachment and unused-cut stretching then embed the witnesses in one induced subcubic grid-tree host, fixed before the controller. Only the starts depend on it.

Layer	Principal conclusion	Proof basis
Parameters	Exact F/M duality; fixed-horizon finite memory	Deductive
Recurrent structure	Tree capacity, sharp all-state period, path phase contact	Deductive
Memoryless and core	Radius law; $F(1) = 4$; $F(2) = F_{\mathcal{T}_{\text{grid},1}}(2) = 7$	Deductive
Path hierarchy	Threshold 8 at two states; a universal controller at four	Finite lower certificate for 8; analytic otherwise
Three-state realization	Direction consistency and one consistent affine-profile system	Deductive iff; scans validate
Obstruction uniformity	Complete test family; protected universal host	Deductive construction; certified selection and coordinate checks

Table 1: The theorem spine. Appendix E separates deductive arguments, necessary finite certificates and implementation checks.

Sections 3–6 establish the structural and low-state results, with the two-state proof in Section 5. Section 7 gives the realization criterion; Sections 8–10 develop complexity and uniform obstructions. Related work and open problems follow. The appendices retain the proof-essential local tables, affine bookkeeping and computational dependencies.

2 Model and extremal parameters

Identify the compass directions with the unit vectors

$$N = (0, 1), \quad E = (1, 0), \quad S = (0, -1), \quad W = (-1, 0), \quad D = \{N, E, S, W\}.$$

Juxtaposition abbreviates a mask, for example $NS = \{N, S\}$. A *maze* is a finite nonempty set $V \subset \mathbb{Z}^2$ whose induced grid graph is connected. Thus every pair of cells of Manhattan distance one is joined by an edge. The mask observed at $x \in V$ is

$$M_V(x) = \{d \in D : x + d \in V\}.$$

We write d_V for graph distance in the induced grid graph and $d_1(x, y) = \|x - y\|_1$ for Manhattan distance; always $d_1(x, y) \leq d_V(x, y)$.

Definition 2.1 (Controller and execution). A controller A consists of a finite nonempty state set Q_A , a distinguished initial state 0, and a deterministic legal transition table

$$\delta_A : Q_A \times (2^D \setminus \{\emptyset\}) \longrightarrow D \times Q_A, \quad \delta_A(q, M) = (d, r), \quad d \in M.$$

The direction is written first and the new state second. Both agents use the same table and the same global compass, start in state 0, and move synchronously. There is no wait action, marking, communication, external clock, or access to coordinates. For $|V| \geq 2$ the one-agent map is

$$\Phi_{A,V}(x, q) = (x + d, r), \quad (d, r) = \delta_A(q, M_V(x)).$$

Starting from cells $x, y \in V$, the two copies iterate this map in parallel from $(x, 0)$ and $(y, 0)$.

Fix a real success radius $\rho \geq 0$. The execution is successful if, for some integer time $t \geq 0$,

$$d_1(X_t, Y_t) \leq \rho.$$

A pair that is already within radius ρ succeeds at time zero. Passing through the same edge in opposite directions creates no additional continuous-time event. Since Manhattan distances on the grid are integral,

$$d_1(X_t, Y_t) \leq \rho \iff d_1(X_t, Y_t) \leq \lfloor \rho \rfloor. \quad (1)$$

A singleton has only a time-zero success and requires no transition at the empty mask. Thus all three parameters below are unchanged when ρ is replaced by $\lfloor \rho \rfloor$.

Let $\mathcal{G}$ denote all finite connected induced grid mazes, $\mathcal{T}_{\text{grid}}$ all induced grid trees, $\mathcal{P}$ all induced grid paths, and $\mathcal{J}$ the class consisting of paths together with induced grid trees having exactly one degree-three vertex and all other degrees at most two. Then

$$\mathcal{P} \subseteq \mathcal{J} \subseteq \mathcal{T}_{\text{grid}} \subseteq \mathcal{G}.$$

Definition 2.2 (Controller-dependent trap size and F). For a nonempty maze class $\mathcal{C}$, let

$$\tau_{\mathcal{C},\rho}(A) = \min \left\{ |V| : V \in \mathcal{C}, \exists x, y \in V \forall t \geq 0, d_1(X_t, Y_t) > \rho \right\},$$

with $\min \emptyset = \infty$. Thus $\tau_{\mathcal{C},\rho}(A) = \infty$ means that the one fixed controller A succeeds from every start pair on every maze in $\mathcal{C}$. For a state budget $s \geq 1$, define

$$F_{\mathcal{C},\rho}(s) = \max_{1 \leq |Q_A| \leq s} \tau_{\mathcal{C},\rho}(A).$$

The controller is chosen first; the adversary then chooses the maze and the starts. The trapping maze may therefore depend on the controller.

For a fixed state budget there are finitely many labelled legal transition tables. Hence the maximum in Definition 2.2 is attained in $\mathbb{N} \cup \{\infty\}$. In particular,

$$F_{\mathcal{C},\rho}(s) = \infty \iff \text{one controller with at most } s \text{ states succeeds on all of } \mathcal{C}. \quad (2)$$

The value ∞ is therefore not an unattained supremum of finite trap sizes.

Definition 2.3 (Finite-horizon state complexity). For $n \in \mathbb{N}_0$, let

$$M_{\mathcal{C},\rho}(n) = \min \left\{ |Q_A| : \forall V \in \mathcal{C}, |V| \leq n, \forall x, y \in V, \exists t \geq 0 : d_1(X_t, Y_t) \leq \rho \right\},$$

again with an empty minimum equal to ∞ . Thus $M_{\mathcal{C},\rho}(n)$ is the least number of states in one controller that succeeds on every legal instance through size n .

The strict threshold relation between the two parameters is immediate but useful enough to record at once.

Theorem 2.4 (*F/M duality*). *For every finite n and every finite state budget $s \geq 1$,*

$$F_{\mathcal{C},\rho}(s) > n \iff M_{\mathcal{C},\rho}(n) \leq s.$$

Proof. By Definition 2.2, $F_{\mathcal{C},\rho}(s) > n$ holds exactly when there is a controller A with $|Q_A| \leq s$ and $\tau_{\mathcal{C},\rho}(A) > n$. The latter inequality says that no maze in $\mathcal{C}$ of size at most n , with any start pair, is a trap for A . Equivalently,

$$\exists A, |Q_A| \leq s : \forall V \in \mathcal{C}, |V| \leq n, \forall x, y \in V, \exists t \geq 0 : d_1(X_t, Y_t) \leq \rho,$$

which is precisely $M_{\mathcal{C},\rho}(n) \leq s$. The equivalence also covers $M = \infty$ and $F = \infty$ under the conventions above. $\square$

Proposition 2.5 (Monotonicities). *The parameters are monotone in the directions shown in Table 2. In addition,*

$$F_{\mathcal{C},\rho}(s) = F_{\mathcal{C},\lfloor \rho \rfloor}(s), \quad M_{\mathcal{C},\rho}(n) = M_{\mathcal{C},\lfloor \rho \rfloor}(n).$$

Proof. Increasing the state budget enlarges the controller set over which F is maximized. Increasing ρ can only turn failures into successes, so for each fixed controller its least trap cannot become smaller; maximizing preserves this direction. If $\mathcal{C}_1 \subseteq \mathcal{C}_2$, the adversary has fewer candidate traps in $\mathcal{C}_1$, so $\tau_{\mathcal{C}_1,\rho}(A) \geq \tau_{\mathcal{C}_2,\rho}(A)$ for every A , and hence the same inequality holds for F .

For M , enlarging the size horizon or the maze class strengthens the universal success requirement and therefore cannot decrease the required number of states; enlarging the success radius weakens that requirement and therefore cannot increase it. The floor identities are exactly (1). $\square$

Change	F	M
$s_1 \leq s_2$	$F_{\mathcal{C},\rho}(s_1) \leq F_{\mathcal{C},\rho}(s_2)$	—
$\rho_1 \leq \rho_2$	$F_{\mathcal{C},\rho_1}(s) \leq F_{\mathcal{C},\rho_2}(s)$	$M_{\mathcal{C},\rho_1}(n) \geq M_{\mathcal{C},\rho_2}(n)$
$\mathcal{C}_1 \subseteq \mathcal{C}_2$	$F_{\mathcal{C}_1,\rho}(s) \geq F_{\mathcal{C}_2,\rho}(s)$	$M_{\mathcal{C}_1,\rho}(n) \leq M_{\mathcal{C}_2,\rho}(n)$
$n_1 \leq n_2$	—	$M_{\mathcal{C},\rho}(n_1) \leq M_{\mathcal{C},\rho}(n_2)$

Table 2: Monotonicities. Larger state budgets and success radii help the controller, whereas enlarging the maze class helps the adversary. For M , increasing the size horizon makes the requirement harder.

For the original model we abbreviate

$$F(s) = F_{\mathcal{G},1}(s), \quad M(n) = M_{\mathcal{G},1}(n).$$

At radius one we also use the fixed-radius shorthands $\tau_{\mathcal{C}}(A) = \tau_{\mathcal{C},1}(A)$ and $F_{\mathcal{C}}(s) = F_{\mathcal{C},1}(s)$, and write $\tau(A) = \tau_{\mathcal{G},1}(A)$. For a fixed controller, Φ_V abbreviates $\Phi_{A,V}$ and x_q denotes the tag (x, q) . An initially noncontacting start-labelled maze whose execution never succeeds is a *trap instance*. These are shorthands for the same model, not new parameters.

$$\mathcal{C}_1 \subseteq \mathcal{C}_2 \implies \tau_{\mathcal{C}_1,\rho}(A) \geq \tau_{\mathcal{C}_2,\rho}(A), \quad F_{\mathcal{C}_1,\rho}(s) \geq F_{\mathcal{C}_2,\rho}(s). \quad (3)$$

Lemma 2.6 (Simultaneous compass transport). *Let $g \in D_4$ be a rotation or reflection of the square grid. Define the transported controller gA by*

$$\delta_{gA}(q, gM) = (gd, r) \quad \text{when} \quad \delta_A(q, M) = (d, r).$$

Transporting the maze and the starts by g carries every execution of A to an execution of gA with the same internal states and the same contact times.

Proof. The induced mask at gx in gV is $gM_V(x)$. Thus $\Phi_{gV}^{gA}(gx, q) = (g(x+d), r)$ whenever $\Phi_V^A(x, q) = (x+d, r)$. Iteration proves the execution identity, and g preserves Manhattan distance. $\square$

The lemma permits compass normalization of a controller-dependent upper witness. It does not permit rotating a maze while leaving a fixed controller unchanged. In the lower proof every orientation is absolute, and the initial state is never exchanged with state 1.

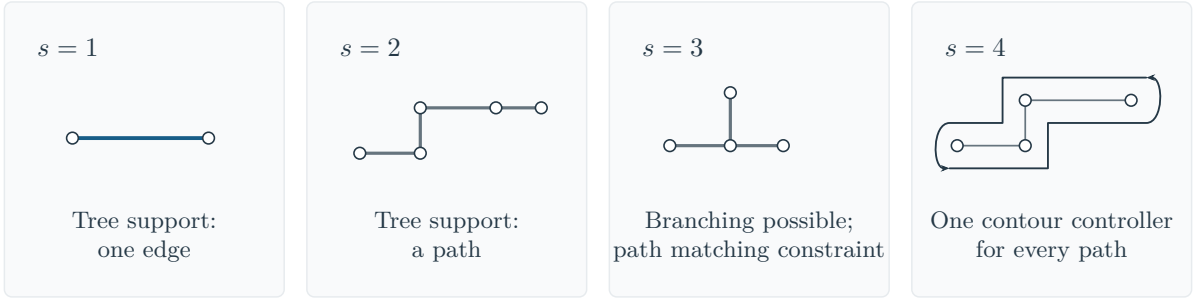


Figure 1: Memory constrains recurrent geometry: the first three panels summarize support capacity, while the last records the universal path controller of Section 6. On three-state paths, matching is necessary; the compatibility theorem in Section 7 supplies the additional realization condition.

3 Recurrent support capacity and period

A *tag* is a pair (v, q) of a maze cell and a controller state. Recurrent tagged cycles are taken in their least period, so their tags are distinct. If

$$\Gamma = ((v_0, q_0), \dots, (v_{L-1}, q_{L-1}))$$

is such a primitive cycle, its *spatial support* is the graph whose vertices are the visited cells and whose edges are the grid edges actually traversed by the cycle. An ambient maze edge is not placed in the support merely because both endpoints are visited.

3.1 Capacity of a tree support

The number of internal states limits how much branching a recurrent tree support can carry.

Theorem 3.1 (Support capacity). *Let A be a compulsory-moving controller with $|Q_A| \leq s$, and let Γ be a primitive recurrent tagged cycle whose spatial support T is a tree. For each used edge e , let k_e be the number of traversals of e in either one of its two directions during one period, and let r_v be the number of tags of Γ lying over v . Then*

$$\boxed{r_v = \sum_{e \ni v} k_e \leq s}, \quad \boxed{|\Gamma| = 2 \sum_{e \in E(T)} k_e}, \quad k_e \geq 1.$$

Consequently

$$\Delta(T) \leq s,$$

and for a square-grid support

$$\Delta(T) \leq \min(s, 4).$$

Proof. Fix a used edge e . Deleting e separates T into components U and W . During one closed period let N_{UW} and N_{WU} be the numbers of crossings of e from U to W and from W to U . The indicator of the current vertex lying in U returns to its initial value after one period, so its net change is zero. Hence

$$N_{UW} = N_{WU} =: k_e.$$

The edge belongs to the used support, so $k_e \geq 1$.

Now fix $v \in V(T)$. Every occurrence of v in the cyclic spatial word has exactly one departure, because motion is compulsory. Conversely every departure from v corresponds to exactly one such occurrence. Along each incident edge e , exactly k_e traversals depart from v . Therefore

$$r_v = \sum_{e \ni v} k_e.$$

Since the tagged cycle is primitive, two occurrences over the same spatial vertex cannot carry the same state: otherwise that tag would repeat before the end of the period. Thus $r_v \leq |Q_A| \leq s$.

Finally,

$$|\Gamma| = \sum_v r_v = \sum_v \sum_{e \ni v} k_e = 2 \sum_e k_e.$$

Since every incident multiplicity is positive,

$$\deg_T(v) \leq \sum_{e \ni v} k_e = r_v \leq s.$$

The grid contributes the independent degree bound four. $\square$

Least period is essential for the capacity inequality: writing a cycle twice doubles r_v without adding states. Theorem 3.1 applies to the unique least tagged period of each recurrent trajectory.

Corollary 3.2 (Low-state tree supports). *For a recurrent cycle with tree support in the compulsory-motion model:*

- (i) *with one state, the support is a single edge;*
- (ii) *with two states, the support is a path;*
- (iii) *with three states, degree-three branching is no longer excluded by state capacity.*

Proof. A recurrent support contains at least one edge. The first two claims follow from the bounds $\Delta(T) \leq 1$ and $\Delta(T) \leq 2$ for a connected tree. For the third, the bound permits degree three, and the next sharpness construction realizes it. $\square$

The degree bound is sharp on induced grid trees for every state budget up to the ambient degree. Let $d = \min(s, 4)$ and take a grid star with centre c and leaves in distinct directions $d_0, \dots, d_{d-1}$. With states $0, \dots, d-1$, define on the used rows

$$\delta(i, \{d_0, \dots, d_{d-1}\}) = (d_i, 0), \quad \delta(0, \{-d_i\}) = (-d_i, i+1 \pmod{d}).$$

Completing the remaining rows arbitrarily gives the primitive cycle

$$c_0, \ell_{0,0}, c_1, \ell_{1,0}, \dots, c_{d-1}, \ell_{d-1,0},$$

whose support has centre degree d . Thus the local transition from path-like recurrence at $s = 2$ to possible branching at $s = 3$ is genuine.

3.2 The sharp path period at every state budget

Here n counts vertices of the used support, not of a larger ambient maze. The following bound is attained by a common mask table, so it is stronger than optimizing the capacity inequalities in a vertex-dependent transition model.

Theorem 3.3 (Sharp maximum path period). *For $s \geq 2$ and $n \geq 3$, the maximum least tagged period over all controllers with at most s states and all fully used n -vertex induced grid paths is*

$$L_{\max}(s, n) = \begin{cases} sn - 2, & n \text{ even}, \\ s(n-1), & n \text{ odd}. \end{cases}$$

For a standalone two-vertex path, $L_{\max}(s, 2) = 2s$. The displayed formula for $n \geq 3$ is also an upper bound for any used path support in a larger maze, regardless of the external masks. Attainment is asserted over geometries and controllers, not on every fixed path.

Proof. Let $k_0, \dots, k_{n-2}$ be the positive edge multiplicities. When $n = 2r + 1$, partition the edges into r consecutive pairs. Each pair has total multiplicity at most s by Theorem 3.1, whence $L \leq 2rs = s(n-1)$. When $n = 2r + 2 \geq 4$, pair the first $2r$ edges in the same way. The remaining edge has multiplicity at most $s-1$, since its neighbour has positive multiplicity. Thus $L \leq 2(rs + s-1) = sn - 2$.

For attainment put $p = s-1$ and take the monotone path with direction word $\text{ENEN} \dots$, beginning with E. Every east edge has multiplicity p ; every north edge has multiplicity one. On its forward

passage replace an east step by $(EW)^{p-1}E$; its backward passage is one west step. Use states $0, 1, \dots, p$ and the following common internal-mask rows:

	0	$1 \leq i < p$	p
ES	(S, p)	(E, i)	$(E, 0)$
NW	$(N, 1)$	$(W, i+1)$	$(W, 0)$

At the left endpoint, of mask E, set $\delta(0, E) = (E, 0)$ if $p = 1$. If $p \geq 2$, use

$$\delta(0, E) = (E, 1), \quad \delta(i, E) = (E, i) \ (2 \leq i < p), \quad \delta(p, E) = (E, 0).$$

For even n the right endpoint has mask W; use $\delta(0, W) = (W, 0)$ and $\delta(i, W) = (W, i+1)$ for $1 \leq i < p$. For odd n it has mask S; use $\delta(1, S) = (S, p)$. All other rows are completed by legal moves.

For an internal east edge $u \rightarrow v$, the forward tags are

$$(u_i, v_i)_{i=1}^{p-1}, \quad u_p, v_0.$$

The last tag moves north in state 1. On the backward contour the next ES vertex in state 0 enters v_p , then u_0 , and continues south. Thus each internal vertex uses all s states exactly once. At the left endpoint replace the first u_1 by the starting tag u_0 ; the specified endpoint rows close the same sequence. The left endpoint uses p tags. The right uses p tags for even n and one for odd n . All tags in the resulting closed word are distinct, giving exactly the claimed least periods. This one table for each s works at every length of this alternating family.

On a standalone edge use $\delta(i, E) = (E, i)$ and $\delta(i, W) = (W, i+1 \bmod s)$ for $0 \leq i < s$. The resulting cycle has $2s$ distinct tags, attaining the capacity bound there as well. $\square$

For one state no recurrent tree support can have $n \geq 3$ vertices. For two states all path multiplicities at $n \geq 3$ are one, but not every prescribed contour is realizable by a shared table. At three states the possible doubled edges form a matching. Section 7 supplies the missing compatibility theorem; the period bound by itself is not a realization criterion.

3.3 Phase contact on a path

The second structural ingredient is independent of automata.

Theorem 3.4 (Phase contact). *Let $p : \mathbb{Z} \rightarrow \mathbb{Z}$ be periodic with period $L \geq 1$ and satisfy*

$$|p(t+1) - p(t)| \leq 1 \quad (t \in \mathbb{Z}).$$

For any phase offsets $\alpha, \beta \in \mathbb{Z}$, there exists $t \in \{0, \dots, L-1\}$ such that

$$\boxed{|p(t+\alpha) - p(t+\beta)| \leq 1.}$$

Proof. Set $d(t) = p(t+\alpha) - p(t+\beta)$. Then $|d(t+1) - d(t)| \leq 2$. Suppose $|d(t)| \geq 2$ for every t . Since d is integer-valued, its sign cannot change between consecutive times: a change from at least 2 to at most -2 , or conversely, would require a jump of magnitude at least four. Periodicity therefore forces one strict sign throughout the whole cyclic period. On the other hand, cyclic reindexing gives

$$\sum_{t=0}^{L-1} d(t) = \sum_{t=0}^{L-1} p(t+\alpha) - \sum_{t=0}^{L-1} p(t+\beta) = 0,$$

a contradiction. $\square$

The theorem permits zero increments, so it remains true for lazy walks with WAIT. It uses no state bound, no grid embedding, no simple tagged-cycle hypothesis, and no assumption that the ambient graph is a tree.

Corollary 3.5 (Same-cycle contact on path support). *Suppose two trajectories of the same controller eventually occupy phase shifts of one recurrent tagged cycle whose spatial support is a path. Then at some integer time the two agents occupy the same or adjacent path vertices.*

Proof. Choose an integer coordinate along the support path. The projection of one period of the recurrent cycle is a periodic integer walk with step size at most one. Once both trajectories lie on that cycle, they are two phase shifts of this word. Theorem 3.4 applies. $\square$

Path support cannot be replaced by a branching-tree hypothesis. An explicit three-state triod with a unique recurrent cycle and two eternally separated state-zero trajectories is given in Appendix D.5.

4 Memoryless controllers at arbitrary radius

The one-state case admits an exact controller-by-controller law for every success radius. Its radius-one specialization recovers the exact one-state threshold.

Theorem 4.1 (Memoryless radius law). *Let A be any one-state controller and let $\rho \geq 0$. For each*

$$\mathcal{C} \in \{\mathcal{P}, \mathcal{J}, \mathcal{T}_{\text{grid}}, \mathcal{G}\},$$

one has

$$\tau_{\mathcal{C},\rho}(A) = F_{\mathcal{C},\rho}(1) = \begin{cases} 2, & 0 \leq \rho < 1, \\ \lfloor \rho \rfloor + 3, & \rho \geq 1. \end{cases}$$

In particular, within each of these four classes every memoryless controller has the same least trap size.

Proof. Write $k = \lfloor \rho \rfloor$. By (1), radius ρ and radius k define exactly the same success event.

First suppose $0 \leq \rho < 1$, so success means coincidence. On a two-cell edge, agents started at opposite endpoints are forced to exchange endpoints forever and never coincide. A singleton has only a time-zero success. Every listed class contains the two-cell path, so the exact value is two.

Now let $k \geq 1$. We first prove a lower bound that does not use memorylessness. If a connected maze V with starts x, y were a trap with $|V| \leq k + 2$, failure at time zero and connectedness would give

$$k + 1 \leq d_1(x, y) \leq d_V(x, y) \leq |V| - 1 \leq k + 1.$$

All inequalities are therefore equalities. A shortest x – y path has $|V| - 1$ edges and visits every vertex. Any extra edge joining nonconsecutive vertices of that path would shorten $d_V(x, y)$, so the whole maze graph is that induced path, with x, y as its endpoints. Compulsory motion forces both first moves inward. If the path is

$$x = v_0, v_1, \dots, v_k, v_{k+1} = y,$$

then after one round the agents are at v_1 and v_k , and

$$d_1(v_1, v_k) \leq d_V(v_1, v_k) = k - 1 \leq k \leq \rho.$$

Thus every controller succeeds on every connected maze of at most $k + 2$ cells.

It remains to give a matching trap for an arbitrary one-state rule. Write $r(M) \in M$ for the chosen direction at mask M and set

$$h = r(\{E, W\}), \quad v = r(\{N, S\}).$$

The unit vectors h and v are perpendicular. On the corner mask $\{h, v\}$ let

$$a = r(\{h, v\}), \quad \{a, b\} = \{h, v\}.$$

Then

$$r(\{a, b\}) = a, \quad r(\{b, -b\}) = b.$$

Consider

$$V_k = \{0, a, b, 2b, \dots, (k + 1)b\},$$

whose path order is $a, 0, b, 2b, \dots, (k + 1)b$. This is an induced path with $k + 3$ cells. Start the two agents at a and kb . The endpoint move from a is forced, the corner at 0 chooses a , an interior b -arm cell chooses b , and the outer endpoint is forced back. Hence

$$(a, kb) \longleftrightarrow (0, (k + 1)b).$$

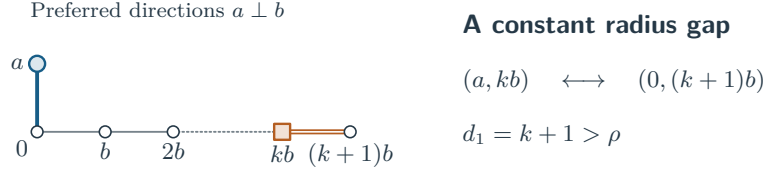


Figure 2: Memoryless trap for $k = \lfloor \rho \rfloor \geq 1$, with starts at a and kb . The axes are preferred directions, not absolute compass labels; the long arm is schematic, with intermediate labels coalescing for $k = 1, 2$.

Both configurations have Manhattan separation $k + 1 > \rho$, so they form a trap. The same witness belongs to all four listed maze classes. $\square$

The class restriction in Theorem 4.1 concerns only the matching upper witness: the small-maze lower bound holds for every class of connected induced grid mazes, while equality follows for any class containing the controller-adapted path V_k .

Corollary 4.2 (The original one-state threshold). *For the original radius-one model,*

$$F_{\mathcal{P},1}(1) = F_{\mathcal{J},1}(1) = F_{\mathcal{T}_{\text{grid}},1}(1) = F(1) = 4.$$

Thus the value four is the first nontrivial instance of the general law $\lfloor \rho \rfloor + 3$, rather than a separate low-state phenomenon requiring its own proof.

5 Exact two-state traps in grid mazes

The general capacity and contact lemmas isolate the recurrent part of the argument, but the exact threshold also requires control of transients and of the distinguished initial state. We prove both bounds without computational assistance.

Theorem 5.1 (Exact two-state trap threshold). *For synchronous radius-one rendezvous in the stated model,*

$$F(2) = F_{\mathcal{J},1}(2) = F_{\mathcal{T}_{\text{grid}},1}(2) = 7.$$

Every controller with at most two states has a trap of at most seven cells that is an induced path or a tree with exactly one degree-three vertex. The controller A^ below succeeds on every connected induced grid maze through six cells and has a seven-cell induced-path trap.*

The upper argument first constrains the table by path witnesses and then uses a junction; it does not assert a paths-only upper bound of seven. The lower argument treats subcubic trees structurally before exhausting the remaining geometries through six cells.

5.1 A seven-cell tree obstruction

We prove a slightly sharper upper bound than required: every controller with at most two states has a trap of size at most seven that is either an induced path or an induced tree with one vertex of degree three. Throughout this section, an action is written $\delta(q, M) = (d, r)$, with direction first. We call a controller a *path survivor* if it succeeds from every pair of state-zero starts on every induced grid path with at most seven cells.

Two elementary observations will certify the traps. If two executions stay in regions whose cross-distances are at least two, they cannot succeed. If instead their regions are merely disjoint, the same conclusion holds when their starts have equal checkerboard color: compulsory synchronous motion preserves equality of their colors, whereas adjacent cells have different colors. In every application below the regions contain the initial positions, or an explicit entry prefix is given.

All compass normalizations act on the controller as well as on the maze. For $g \in D_4$, define

$$(g\delta)(q, gM) = (gd, r) \quad \text{when } \delta(q, M) = (d, r).$$

Induction on time conjugates the complete tagged executions under $(x, q) \mapsto (gx, q)$. Manhattan distance and initial state zero are preserved. In particular, a trap for the transported controller gives a trap for the original controller by applying g^{-1} .

5.1.1 Straight-axis rigidity

Lemma 5.2 (Axis reduction). *For a path survivor, one simultaneous compass symmetry makes the straight rows*

$$\begin{aligned}\delta(0, \text{NS}) &= (\text{S}, f_v(0)), & \delta(1, \text{NS}) &= (\text{N}, f_v(1)), \\ \delta(0, \text{EW}) &= (\text{W}, f_h(0)), & \delta(1, \text{EW}) &= (\text{E}, f_h(1)).\end{aligned}$$

Each map $f = f_v, f_h$ is a constant map C_0, C_1 or the identity I . Write L_q for the next-state bit at the negative endpoint of an axis and R_q for that at its positive endpoint. The necessary restrictions are

f	endpoint restrictions	
C_0	$L_0 = 1, R_0 = 0,$	(5.2a)
I	$L_0 = 1, (R_0, R_1) \neq (1, 1),$	
C_1	$L_1 = 1, (R_0, R_1) \neq (1, 1).$	

All unlisted endpoint bits remain arbitrary.

Proof. For perpendicular unit directions d, e , the seven cells

$$U(d, e) = \{0, e, 2e, d, 2d, 2e + d, 2e + 2d\}$$

form an induced path with two parallel two-edge arms pointing in direction d . If both straight rows on an axis choose d , starts at d and $2e + d$ stay on translated terminal edges, regardless of all next-state bits. Their distance is always two. Thus the straight directions are opposite, and independent coordinate reflections give the displayed normalization without exchanging the states.

Index a straight path by increasing coordinates. If $f = (1, 0)$, the six-cell path $0, \dots, 5$ with starts $2, 4$ gives the two tagged cycles $(2_0, 1_1)$ and $(4_0, 3_1)$; the agents remain two cells apart. If $f = C_0$ and $R_0 = 1$, the five-cell path $0, \dots, 4$ with starts $0, 4$ confines the right agent to $(4_0, 3_1)$ and the left to $0, 1, 2$. Indeed the left word is $(0_0, 1_0)^\infty$ when $L_0 = 0$, and $(0_0, 1_1, 2_0, 1_0)^\infty$ when $L_0 = 1$. The regions are disjoint and the starts have equal color. If $f = C_1$ and $L_1 = 0$, use the same line with starts $1, 3$. The left cycle is $(1_0, 0_1)$; the right prefix $3_0 \rightarrow 2_1 \rightarrow 3_1$ enters the closed region $2, 3, 4$. At its positive endpoint, either return bit keeps it in that region. Again equal colors exclude adjacency.

Two terminal-edge tests finish the restrictions. If $L_{f(0)} = 0$, an inner state-zero start has cycle $I_0 \rightarrow E_{f(0)} \rightarrow I_0$ at a negative terminal edge. Two copies in $U(d, e)$, with d negative and inner starts, give a trap. If $R_0 = R_{f(1)} = 1$, an endpoint start at a positive terminal edge has word $E_0 \rightarrow I_1 \rightarrow E_{f(1)} \rightarrow I_1 \rightarrow \dots$. Clone these words in the positive-pointing $U(d, e)$, with endpoint starts. Both constructions keep the agents two units apart. The four possible two-bit maps, together with these exclusions, give exactly the necessary restrictions displayed above. $\square$

For singleton masks write $\delta(q, \text{N}) = (\text{N}, n_q)$, and similarly e_q, s_q, w_q . Thus $(L, R) = (n, s)$ vertically and (e, w) horizontally. For perpendicular directions u, v , put

$$L(u, v; a, b) = \{C, u_1, \dots, u_a, v_1, \dots, v_b\}, \quad C = (0, 0), \quad d_j = jd.$$

This is an induced path with $a + b + 1$ cells.

Lemma 5.3 (Reset pocket). *Suppose a straight state map resets to κ , and let d be the direction selected by state κ . On a terminal arm C, d_1, d_2, d_3 , a state-zero start at d_2 never reaches C .*

Proof. The tagged set $\{(d_1)_\kappa, (d_2)_0, (d_2)_1, (d_3)_\kappa\}$ is closed. At d_2 , either action reaches d_1 or d_3 in state κ . From $(d_1)_\kappa$ the next move goes outward to $(d_2)_\kappa$; the endpoint $(d_3)_\kappa$ returns to d_2 in either state. The set contains the initial tagged position. $\square$

Lemma 5.4 (Straight-axis rigidity). *For a path survivor in the frame of Lemma 5.2, both straight state maps are the identity. Consequently*

$$n_0 = e_0 = 1, \quad (s_0, s_1), (w_0, w_1) \neq (1, 1).$$

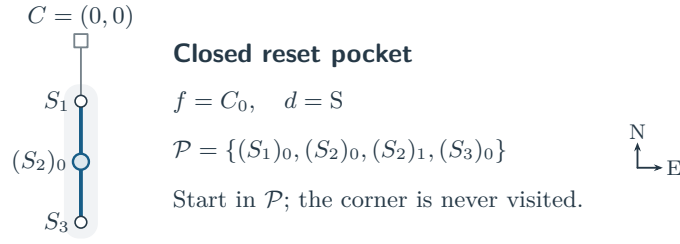


Figure 3: The closed tagged pocket for reset state 0 and direction S; unshown neighbours of C do not affect it. Other reset directions transport both the geometry and the controller directions together.

Proof. If both axes reset, choose their outward reset directions d_v, d_h . On $L(d_v, d_h; 3, 3)$ start at $(d_v)_2, (d_h)_2$. Lemma 5.3 confines the agents to different three-cell arms, whose cross-distances are at least two. This treats all four choices of the two reset bits.

If exactly one axis resets, interchange the axes simultaneously to make it vertical. The two remaining possibilities are C_0/I and C_1/I . Lemmas A.1 and A.2 give explicit path obstructions for each in Appendix A.1. Their proofs use reset pockets, closed arms, and short transition words; all endpoint bits allowed by Lemma 5.2 are retained. Thus neither mixed possibility can survive. The endpoint conclusions now follow from Lemma 5.2. $\square$

5.1.2 Corner compatibility and a nine-entry pattern

Lemma 5.5 (Corner arrival). *Use the normalized straight directions of Lemma 5.2. Let C, I, E be a terminal arm of two edges, with I a straight cell and direction d from C into the arm. Every arrival at C from this arm has state $a_d = f(q_{-d})$, where state q_{-d} selects the straight direction toward C . If the corner action in state a_d points back into the arm, a state-zero start at I remains in C, I, E . For identity straight maps,*

$$a_N = a_E = 0, \quad a_S = a_W = 1.$$

Proof. The last move $I \rightarrow C$ gives the arrival formula. Endpoints return to I , straight moves stay in the arm or reach C in that state, and the specified corner row sends such an arrival back to I . This proves closure from the stated initial position, with no restriction on endpoint or corner output bits. $\square$

Call the condition in this lemma a *locked arm*. Two corners two units apart, joined by their middle cell, can carry perpendicular two-edge arms on either side of the connector. If both arms are locked, starts at their inner cells stay in parallel rows or columns two units apart. Each construction is an induced seven-cell path.

Lemma 5.6 (Corner compatibility). *After one further diagonal reflection, a path survivor satisfies*

$$\begin{aligned} \delta(0, NE) &= (E, 1), & \delta(0, ES) &= (S, 0), \\ \delta(1, ES) &= (E, 1), & \delta(1, SW) &= (S, 0). \end{aligned} \tag{5.6a}$$

Proof. The diagonal reflection exchanges the axes, preserves the four identity straight rows, and lets us choose the direction of $\delta(0, NE)$ to be E. Its east arm is locked. The following three paired-arm obstructions force the other directions. First put an NE corner at $(0, 0)$ with east arm and an SW corner at $(0, 2)$ with west arm, joined through $(0, 1)$. If SW in state 1 chose west, both arms would be locked. Hence SW in state 1 chooses south. Next replace the upper corner by ES with an east arm; if ES in state 0 chose east, both east arms would be locked. Thus ES in state 0 chooses south. Finally use $U(S, E)$: its two corners have masks ES and SW. The SW south arm is locked, so ES in state 1 cannot also choose south. It must choose east. Starts in all three constructions are the inner cells of the two arms.

These direction conclusions leave four output bits. The four six-cell obstructions in Lemma A.3 force them to be, respectively, 1, 0, 1, 0. Each argument uses disjoint regions and equal-color starts, and none assumes any of the other three bit conclusions. $\square$

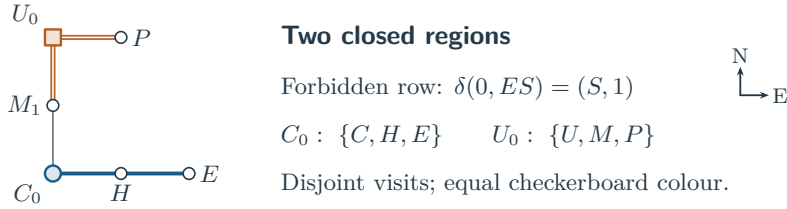


Figure 4: Corner obstruction under the stated forbidden row and preceding rigidity hypotheses. The U -started execution reaches M only in state 1; disjointness excludes coincidence and parity excludes adjacency.

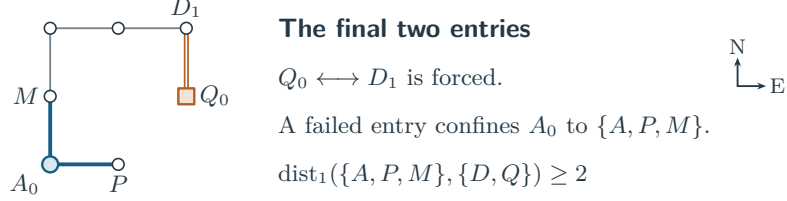


Figure 5: The final seven-cell path, anchored at $A = (0, 0)$. If either remaining W9 entry fails, the left execution stays separated from the forced right-hand tagged two-cycle, including the transient.

Lemma 5.7 (Nine-entry pattern). *Every path survivor admits a simultaneous compass frame in which the following entries hold:*

M	q	$\delta(q, M)$	M	q	$\delta(q, M)$
N	0	(N, 1)	NE	1	(N, 1)
E	0	(E, 1)	ES	0	(S, 0)
W	1	(W, 1)	ES	1	(E, 1)
EW	1	(E, 1)	SW	1	(S, 0)
NE	0	(E, 1)			

(5.7a)

No junction entry is prescribed.

Proof. Only $w_1 = 1$ and $\delta(1, NE) = (N, 1)$ remain to be proved. Use the path, in endpoint order,

$$(1, 0), (0, 0), (0, 1), (0, 2), (1, 2), (2, 2), (2, 1),$$

with state-zero starts $A = (0, 0)$ and $Q = (2, 1)$. Put $P = (1, 0)$, $M = (0, 1)$, and $D = (2, 2)$. The right agent has cycle $Q_0 \rightarrow D_1 \rightarrow Q_0$. The left starts $A_0 \rightarrow P_1$. If $w_1 = 0$, it returns to A_0 and repeats. If $w_1 = 1$, it reaches A_1 . Should the NE state-one row choose east, A, P is closed; should it be $(N, 0)$, its only extra excursion is $A_1 \rightarrow M_0 \rightarrow A_0$. In every bad case the left stays in A, P, M , all at distance at least two from Q, D . Thus the two remaining entries are forced, and the other seven come from Lemmas 5.4 and 5.6. $\square$

5.1.3 Completion at a junction

The last step uses only the nine entries of Lemma 5.7; all other rows remain arbitrary.

Lemma 5.8 (Junction completion). *Every legal completion of the nine-entry pattern has a trap on one of the two seven-cell induced trees G_1, G_2 below.*

Proof. Write $\delta(1, E) = (E, \epsilon)$ and $u = \delta(0, NSW)$, $v = \delta(1, NSW)$. The common cells of the two trees are

$$A = (0, 1), \quad B = (1, 0), \quad J = (1, 1), \quad U = (1, 2), \quad V = (2, 2).$$

For G_1 add $D = (2, 0), F = (3, 0)$ and start at A, B . For G_2 add $Q = (3, 1), R = (3, 2)$ and start at B, Q . All starts have state zero. The edge sets are

$$\begin{array}{l|l} G_1 & JA, JB, JU, UV, BD, DF, \\ G_2 & JA, JB, JU, UV, VR, RQ. \end{array}$$

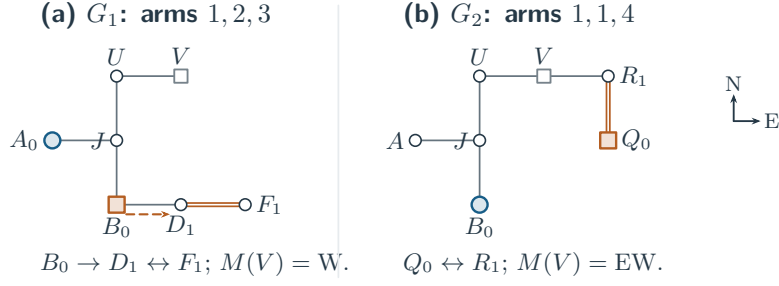


Figure 6: Junction completions with $J = (1, 1)$ and $M(J) = \text{NSW}$. The highlighted right-hand shuttle is fixed in each selected case; the other execution is governed by the exact alternatives in the proof.

Direct coordinate comparison gives no other unit-distance pair. Each tree has one degree-three vertex J , with mask NSW . Choose G_2 precisely when

$$v = (S, 0) \quad \text{or} \quad [u = (S, 0) \text{ and } v \in \{(N, 0), (W, 1)\}]. \quad (5.8a)$$

Choose G_1 otherwise.

On G_1 , the B-started agent follows $B_0 \rightarrow D_1 \rightarrow F_1 \rightarrow D_1 \rightarrow \dots$. The A-started agent begins $A_0 \rightarrow J_1$. Its relevant nonjunction transitions are

$$\begin{aligned} A_0 \rightarrow J_1, \quad A_1 \rightarrow J_\epsilon, \quad B_1 \rightarrow J_1, \\ U_0 \rightarrow J_0, \quad U_1 \rightarrow V_1, \quad V_1 \rightarrow U_1. \end{aligned}$$

If neither u nor v is $(S, 0)$, these transitions and the allowed junction departures close the tagged set

$$\{A_0, A_1, J_0, J_1, B_1, U_0, U_1, V_1\}.$$

The remaining G_1 cases have $u = (S, 0)$ and $v \in \{(N, 1), (S, 1), (W, 0)\}$. They send the A-started agent, respectively, to the edge UV , to the state-one cycle on JB , or to $J_1 \leftrightarrow A_0$. It never queries J_0 , so the row u is harmless. Thus from time one the two agents lie on disjoint regions, one on DF and the other outside it. At time zero they are distinct and have equal checkerboard color, which excludes adjacency at every time. Hence G_1 is a trap in every case where it was selected.

On G_2 the Q-started cycle is always $Q_0 \leftrightarrow R_1$. The selection condition gives exactly the following B-started words:

$$\begin{array}{l|l} v = (S, 0) & (B_0, J_1)^\infty \\ u = (S, 0), v = (N, 0) & (B_0, J_1, U_0, J_0)^\infty \\ u = (S, 0), v = (W, 1), \epsilon = 0 & (B_0, J_1, A_1, J_0)^\infty \\ u = (S, 0), v = (W, 1), \epsilon = 1 & B_0, (J_1, A_1)^\infty. \end{array}$$

Every indicated move is fixed by the nine entries or the displayed choice of u, v, ϵ . At each integer time the distance from the other agent is exactly three, as follows directly from the coordinates. This includes all transients and finishes the proof. $\square$

Proposition 5.9 (Upper bound in a narrow tree class). *Every controller with at most two states has a trap with at most seven cells that is either an induced path or an induced tree with exactly one vertex of degree three and no larger degree. In particular, $F(2) \leq 7$ and $F_{\text{grid trees}}(2) \leq 7$.*

Proof. A two-state controller either already has a path trap of size at most seven, or is a path survivor. In the second case apply Lemma 5.7, then Lemma 5.8, and transport the resulting tree and starts back by the inverse compass symmetry. A one-state controller can be extended by a legal unused second state without changing any execution from its distinguished initial state. The two-state argument therefore covers it as well. $\square$

5.2 An extremal controller and its seven-cell trap

Table 3 specifies all thirty entries of A^* , with direction first and updated state second.

Mask type	Mask	State 0	State 1
End	N	(N, 0)	(N, 1)
	E	(E, 1)	(E, 1)
	S	(S, 1)	(S, 1)
	W	(W, 1)	(W, 0)
Straight	NS	(S, 1)	(S, 1)
	EW	(W, 0)	(E, 1)
Corner	NE	(N, 0)	(E, 1)
	ES	(S, 1)	(E, 1)
	SW	(W, 0)	(S, 1)
	NW	(N, 1)	(W, 0)
Junction	NES	(S, 1)	(E, 1)
	NEW	(W, 0)	(E, 1)
	NSW	(S, 0)	(S, 0)
	ESW	(S, 1)	(S, 1)
Full	NESW	(E, 0)	(S, 1)

Table 3: The extremal controller A^* . Every direction belongs to its mask. The distinguished initial state is 0.

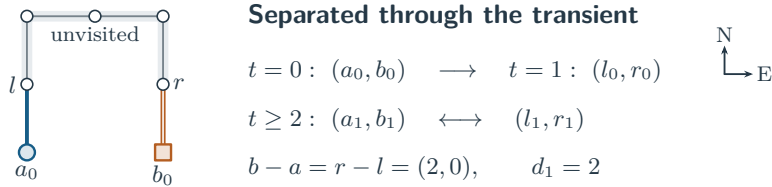


Figure 7: The seven-cell U-trap for A^* , with $a = (0, 0)$ and $b = (2, 0)$. The upper connector is never visited; from time 2 both executions shuttle in state 1 on the highlighted edges.

On a vertical straight segment both states move south and reset to 1. On a horizontal straight segment state 0 moves west and state 1 moves east, each preserving its state. The masks NES and NEW behave like ES and EW, respectively, so their extra northern arms are never selected. The remaining three-way masks also avoid north. These local asymmetries will determine the possible recurrent supports.

Lemma 5.10 (Seven-cell sharpness witness). *The controller A^* has a trap on the induced path*

$$V_U = \{(0, 0), (0, 1), (0, 2), (1, 2), (2, 2), (2, 1), (2, 0)\},$$

with state-0 starts at $(0, 0)$ and $(2, 0)$.

Proof. The bottom cells have mask N, and their northern neighbours have mask NS. The entire execution through the first repeated product state is

t	first agent	second agent	distance
0	$(0, 0)_0$	$(2, 0)_0$	2
1	$(0, 1)_0$	$(2, 1)_0$	2
2	$(0, 0)_1$	$(2, 0)_1$	2
3	$(0, 1)_1$	$(2, 1)_1$	2
4	$(0, 0)_1$	$(2, 0)_1$	2

Time 4 repeats time 2. Determinism gives separation two forever, including the transient. The product preperiod and period are both exactly two. $\square$

For the lower bound, we prove that A^* succeeds through six cells. Tagged recurrence matters: visits to one cell can carry different states, and separated eventual supports may still force transient contact.

5.3 Recurrent supports on trees

We first separate elementary contact principles from the special features of A^* . A *tagged cycle* means a simple directed cycle of Φ_V , and its *support* consists of the spatial vertices and edges used by that tagged cycle. In particular, the support does not include an ambient edge merely because both its endpoints occur. A tagged cycle is *reachable* if some initial tag $(x, 0)$, $x \in V$, eventually enters it. We call a support *long* if it has at least two edges.

Lemma 5.11 (Two-state path support). *For any compulsory-moving controller with at most two states, a primitive tagged cycle whose support is a tree has nonempty path support. If that path has $a \geq 2$ edges, the tagged cycle traverses each edge once in each direction and has period $2a$. On a one-edge support the tagged period is two or four.*

Proof. The path-support conclusion is Corollary 3.2. If the path has at least two edges, an internal vertex v has two incident support edges. Theorem 3.1 gives

$$2 \leq \sum_{e \ni v} k_e = r_v \leq 2,$$

so both incident multiplicities equal one. Propagation along the connected path gives $k_e = 1$ on every edge, and Theorem 3.1 then gives period $2a$. On a single edge, $r_v = k_e \leq 2$, so $k_e \in \{1, 2\}$ and the period is 2 or 4. $\square$

Lemma 5.12 (Contact on one tagged cycle). *If two trajectories eventually enter the same tagged cycle and its support is a path, they achieve distance-one contact. Their initial tags and entry times may be arbitrary, and the ambient maze need not be a tree.*

Proof. This is Corollary 3.5. $\square$

Lemma 5.13 (Contact through an edge support). *Suppose one eventual tagged cycle is supported on an edge uv , and the other eventual tagged cycle visits u or v . The trajectories achieve distance-one contact, independently of their entry times and phases.*

Proof. After both transients have ended, the first agent is always at u or v . At any subsequent visit of the second agent to the shared endpoint, their distance is at most one. Such visits recur. $\square$

Lemma 5.14 (Order inversion). *Two synchronous nearest-neighbour walks on a path cannot reverse their strict spatial order without first achieving distance-one contact. In particular, contact occurs if their eventual disjoint support intervals occur in the opposite order from their initial positions.*

Proof. In path coordinates the difference of their positions changes by at most two at each step. As long as its absolute value is at least two, its sign cannot reverse. This proves the assertion, including the whole transient. $\square$

The support hypothesis in Lemma 5.12 and the edge hypothesis in Lemma 5.13 will be established before either lemma is applied. Neither contact nor equality of spatial positions requires equality of the internal states.

5.3.1 A north-boundary invariant

For a support in a tree, call an edge from a support vertex to a vertex outside the support a *boundary edge*, directed away from the support. All directions in the following statements refer to the fixed compass of Table 3.

Lemma 5.15 (Exclusion of a full vertex). *On an induced grid tree, a long tagged cycle of A^* contains no vertex with mask $NE\!SW$.*

Proof. Suppose first that a full vertex p occurs in state 1. Its next tag is s_1 , where s is its southern neighbour. If s is internal to the support, one of its two support exits is north, and its state-1 exit must continue away from p . The only masks with two distinct exits including north are NE and NW . Accordingly s has an eastern or western neighbour. That neighbour, s , p , and the corresponding eastern or western neighbour of p form a unit square, contradicting inducedness and the tree hypothesis. If s is a support endpoint, returning north from state 1 requires mask N and returns to p_1 . This closes a two-tag cycle, contradicting long support.

A full vertex occurring only in state 0 must be a support endpoint, by Lemma 5.11. Its eastern neighbour is internal and is entered in state 0. To continue away from p in that state and return west in the other state, this neighbour must have mask NW . Its northern neighbour and the northern neighbour of p again complete a unit square. This excludes the remaining case. $\square$

Lemma 5.16 (North boundary). *For A^* on any finite induced grid tree, every boundary edge of a long tagged cycle points north.*

Proof. An internal support vertex uses both states and selects its two support edges. The masks with distinct selected directions are $NE, ES, NW, EW, SW, NES, NEW, NESW$. Lemma 5.15 excludes $NESW$. The mask NW is also impossible internally: to pass from its western support neighbour towards its northern support neighbour, the vertex must be entered in state 0. Every eastward move from a non-full vertex arrives in state 1, whereas the predecessor cannot be full.

The remaining internal possibilities are displayed below. Here d/r denotes direction d and new state r , in the convention $\delta(q, M) = (d, r)$. The last column lists only neighbours outside the two selected support edges.

Internal mask	From 0	From 1	Possible boundary port
EW, NEW	W/0	E/1	N , for NEW
NE	N/0	E/1	none
SW	W/0	S/1	none
ES, NES	S/1	E/1	N , for NES

It remains to check support endpoints; no global assumption about the shape of the path is needed. Let c be an endpoint and v its adjacent internal support vertex.

If v is east of c , its westward return enters c in state 0. To avoid immediately repeating that return, the eastward move from c must arrive at v in state 1. Only an E leaf, used in state 0, does so. If v is west of c , its eastward return enters c in state 1; the westward return from c must arrive at v in state 0. Exactly W and NW in state 1 have this behaviour.

If v is north of c , its southward move arrives at c_1 . The only state-1 northward move is from an N leaf and arrives in state 1. The internal neighbour cannot be SW , since this would repeat the same two tags immediately; it is ES or NES . Finally, if v is south of c , then v has mask NE , whose northward move enters c_0 . The southward return must arrive at v in state 1. The candidate masks at c are S, NS, ES, NES, ESW . Each of the last three requires an eastern neighbour; together with c, v , and the eastern support neighbour of v , it forms a unit square. Only S and NS remain.

These four directional equations give the complete endpoint table.

Endpoint mask	Tag	Direction to support	Boundary port
E	0	E	none
N	1	N , towards ES/NES	none
S, NS	0	S , towards NE	N , for NS
W, NW	1	W	N , for NW

Every boundary port in the two tables points north. $\square$

Both preceding lemmas are independent of maze size and of maximum degree. Inducedness is used when a required fourth side turns a local configuration into a spatial square.

5.3.2 Short supports and the six-cell budget

Lemma 5.17 (Edge profiles). *Every edge-supported tagged cycle of $A^\star$, on any maze, has period two and belongs to one of the five rows of Table 4. On a connected induced tree with more than two cells and maximum degree at most three, these reduce to the five port profiles in Table 5. None has a south-pointing boundary edge.*

Proof. A four-tag horizontal cycle would require both states at the western endpoint to move east. Only an E leaf does this, and both moves arrive in state 1, which precludes four distinct tags. A four-tag vertical cycle requires an N leaf at the lower endpoint. At its upper endpoint both states must move south, and every such mask resets the state to a constant. This again precludes four distinct tags.

For period two, solve the two endpoint equations in the semantic table. For example, a lower state-0 tag that moves north as 0 has mask N or NE ; the upper tag returns south as 0 precisely with mask NSW . This gives the first row. A lower 0 that arrives north as 1 requires NW , and its upper 1 must return south as 0, again with NSW . A lower 1 moving north requires N , and its upper 1 must move south as 1, giving the third row. Horizontally, an eastward arrival as 0 requires the full western vertex in state 0, followed by a state-0 westward return; an eastward arrival as 1 followed by return to western state 0 requires the western E leaf and an eastern W or NW tag in state 1. A western tag in state 1 would require a westward arrival as 1, which is available only from an eastern W leaf in state 0; no state-1 eastward move can return to that eastern tag in state 0. These are all possible state pairs and directions.

Orientation; endpoint tags	Western/lower mask	Eastern/upper mask
Vertical; 0, 0	N, NE	NSW
Vertical; 0, 1	NW	NSW
Vertical; 1, 1	N	$S, NS, SW, ESW, NESW$
Horizontal; 0, 0	$NESW$	EW, NEW, SW
Horizontal; 0, 1	E	W, NW

Table 4: All local edge-supported tagged cycles of $A^\star$. The two tags are ordered west–east or south–north.

The second row forces western neighbours at both endpoints and therefore a unit square; it is excluded on an induced tree. The fourth row and the full-mask option of the third are excluded by the degree bound. A horizontal E – W pair or a vertical N – S pair forms an entire two-cell connected component, excluded by the size assumption. The remaining possibilities are exactly Table 5. $\square$

Profile	Tail/head masks	Tagged cycle	Boundary ports	Cells required
H	$E; NW$	u_0, h_1	N at h	1
D_N	$N; NS$	u_1, h_1	N at h	1
D_W	$N; SW$	u_1, h_1	W at h	1
D_{EW}	$N; ESW$	u_1, h_1	E, W at h	2
Z	N or $NE; NSW$	u_0, h_0	N, W at h , and E at u if u has mask NE	2 or 3

Table 5: The five edge profiles on subcubic induced trees with more than two cells. The tail u lies west of the head h in H , and south of it otherwise. The last column counts required cells outside the two-cell support, before a connecting path is chosen.

Lemma 5.18 (Disjointness of recurrent supports). *Distinct tagged cycles of $A^\star$ on a subcubic induced grid tree have disjoint spatial supports.*

Proof. An internal support vertex uses both tags and cannot occur on a second tagged cycle. A shared spatial vertex must therefore be an endpoint of both supports, in opposite states. The endpoint table in the proof of Lemma 5.16 allows no mask in both endpoint states, so two long supports cannot intersect.

Comparing that table with Table 5 leaves two possible mask coincidences for a long and an edge support. An NS endpoint in state 0 could coincide with a D_N head in state 1, but its southern

neighbour would have to be both an internal NE vertex and an N leaf. An N endpoint in state 1 could coincide with the lower Z endpoint in state 0, but its northern neighbour would have to be both ES/NES and NSW . Neither is possible. For two edge supports the only opposite-state coincidence is an N tail in states 0 and 1; its upper neighbour would require mask NSW for the former and NS, SW , or ESW for the latter. Thus it is impossible as well. The two-cell maze has only its single edge cycle, so it creates no omitted exception. $\square$

Lemma 5.19 (Uniqueness of a long support through six). *If $|V| \leq 6$, V is a subcubic induced grid tree, and A^* has a long tagged cycle on V , it has no other tagged cycle.*

Proof. By Lemma 5.18, another support would be disjoint. A long support has only north boundary edges, whereas any other support has no south boundary edge, by Lemmas 5.16 and 5.17. Direct adjacency between the supports is therefore impossible. Their unique connecting path has an intermediate vertex.

Two long supports require at least $3 + 3 + 1 = 7$ vertices. A long support of at least four vertices paired with an edge support also requires at least seven. The only remaining budget is a three-vertex long support, a two-vertex edge support, and one connecting vertex. An edge profile D_{EW} or Z requires another branch off this minimal union, giving a seventh vertex. For H or D_N , both boundary ports are north, so the two-step connector would be NS and return to its starting point. For D_W , it must be NE . The southern support leaf of the D_W head is then adjacent to the starting support vertex. It either overlaps the long support or closes a unit square. Every case is excluded. $\square$

Lemma 5.20 (The four short-support configurations). *If a subcubic induced grid tree of at most six cells contains two edge-supported tagged cycles of A^* , then, up to translation, it is exactly one of the four six-cell sets in Table 7. Each has exactly two tagged cycles.*

Proof. The supports are disjoint by Lemma 5.18. Let their unique connector have k edges, excluding the two support edges themselves. The union already has $k + 3$ vertices. A required port off this union costs a further distinct vertex: identifying that neighbour with the union would create a spatial cycle. A connector attached to a D_{EW} head has one such extra cost. At a Z head the cost is one; when it attaches at the eastern port of the lower Z vertex, the cost is two. The other profiles have no extra cost after their unique boundary port has been selected.

The resulting fifteen unordered comparisons are recorded in Table 6. Connector words are read from the first named profile to the second. Exchanging their names only relabels the connector endpoints and does not transform compass directions.

Profile pair	Budget	Outcome
H, H	$k \leq 3$	North–north obstruction
H, D_N	$k \leq 3$	North–north obstruction
D_N, D_N	$k \leq 3$	North–north obstruction
D_W, D_W	$k \leq 3$	West–west obstruction
H, D_W	$k \leq 3$	Only NEE ; gives P_H
D_N, D_W	$k \leq 3$	Only NEE ; gives P_N
H, D_{EW}	$k \leq 2$	Square or overlapping supports
D_N, D_{EW}	$k \leq 2$	Square
D_W, D_{EW}	$k \leq 2$	Only WW ; gives P_E
D_{EW}, D_{EW}	$k \leq 1$	Adjacent southern leaves close a square
H, Z	$k \leq 2$ at head; ≤ 1 below	Square at head; incompatible lower-port directions
D_N, Z	$k \leq 2$ at head; ≤ 1 below	Square at head; incompatible lower-port directions
D_W, Z	$k \leq 2$ at head; ≤ 1 below	Square at head; direct W into the lower vertex gives P_Z
D_{EW}, Z	$k \leq 1$ at head	Square; lower attachment exceeds the budget
Z, Z	$k \leq 1$ between heads	No compatible direct ports; other attachments exceed the budget

Table 6: All fifteen pairings of the five edge profiles. The proof gives the direction and square obstruction behind every rejected row. No compass orientations are identified.

Here are the complete route arguments. Two north ports require a connector beginning N and ending S . At length at most three, the only simple possibilities are NES, NWS ; their endpoint heads

are horizontally adjacent, so the induced edge closes a square. Shorter possibilities revisit a vertex. For two west ports the corresponding words are WNE, WSE , and the vertically adjacent heads again close a square. A north port joined to a west port requires final step E . The length-two word NE closes a square through the second head's southern leaf. At length three, NNE closes the same square one level higher, and NEE is the only other simple word. This proves the first six rows.

A D_{EW} attachment requires a horizontal final step. From a north port, the budget allows only NE or NW ; the southern support leaf of the second head is adjacent to the first head, closing a square or overlapping a support. From a west port, a one-step connection makes the southern leaves adjacent and closes a square; among two-step words, WE repeats the starting vertex and WW survives. Two D_{EW} heads have only the already excluded one-step geometry. This proves rows seven through ten.

A Z head accepts outside neighbours north and west, so a connector entering it ends in S or E . From a north port with $k \leq 2$, the only simple word is NE , which closes a square through the lower Z support vertex. From a west port, WE repeats the start, and WS closes a square through the first head's southern leaf. Direct westward entry would approach the forbidden eastern side of the Z head. At the lower Z endpoint the outside port is east; the two extra head neighbours force $k \leq 1$, and the one-step westward connection matches only a D_W head. A $D_{EW}-Z$ head connection has only one step available: its compatible horizontal direction again leaves adjacent southern support vertices and hence a square. Connecting instead to the lower Z vertex would need at least seven cells. Finally, a direct $Z-Z$ head connector would have to begin in N or W and end in S or E , which is impossible in one step; any use of a lower port exceeds the budget. This proves the remaining five rows.

Pattern	Cell set
P_N	$\{(0, -1), (0, 0), (0, 1), (1, 1), (2, 1), (2, 0)\}$
P_E	$\{(-1, 0), (0, -1), (0, 0), (1, 0), (2, 0), (2, -1)\}$
P_Z	$\{(0, 0), (0, 1), (0, 2), (-1, 1), (1, 0), (1, -1)\}$
P_H	$\{(-1, 0), (0, 0), (0, 1), (1, 1), (2, 1), (2, 0)\}$

Table 7: The four possible subcubic six-cell trees with two tagged cycles. Coordinates retain absolute orientation.

The surviving words and required port cells give exactly the four displayed sets. Their masks satisfy the stated edge profiles, so both tagged cycles exist, and no additional branch fits. A third tagged cycle would have disjoint support and use only the two vertices outside the first two supports. In P_N, P_H , those two vertices have masks ES, EW , which do not form any row of Table 4. In P_E, P_Z , they are not adjacent. Thus there is no third tagged cycle, including one supported on connector vertices. $\square$

5.3.3 Initial-state gates and transient contact

Two spatially separated tagged cycles do not by themselves constitute a trap. We must determine which state-0 roots reach them and whether the approach trajectories have already contacted.

Lemma 5.21 (Three closed state-0 gates). *In each of P_N, P_E, P_Z , every state-0 root reaches the same tagged cycle.*

Proof. In all three patterns the right cycle is the state-1 vertical edge at a D_W head r and its southern N leaf b . However, r_0 moves west as 0, and b_0 first enters r_0 . These are escape routes from the right support.

For P_N , put $\ell = (0, 0)$, $a = (0, -1)$, $p = (0, 1)$, $m = (1, 1)$, $r = (2, 1)$, and $b = (2, 0)$. The NS head ℓ sends both states south into the cycle (ℓ_1, a_1) . Every state-0 root reaches it, as witnessed by

$$a_0 \rightarrow \ell_0 \rightarrow a_1, \quad p_0 \rightarrow \ell_1, \quad m_0 \rightarrow p_0, \quad r_0 \rightarrow m_0, \quad b_0 \rightarrow r_0.$$

The potentially east-going tag p_1 is never reached by a state-0 root.

For P_E , put $\ell = (0, 0)$, $a = (0, -1)$, $w = (-1, 0)$, $m = (1, 0)$, $r = (2, 0)$, and $b = (2, -1)$. The ESW head ℓ sends both states to a_1 , and the left cycle is (ℓ_1, a_1) . The remaining roots satisfy

$$w_0 \rightarrow \ell_1, \quad a_0 \rightarrow \ell_0, \quad m_0 \rightarrow \ell_0, \quad r_0 \rightarrow m_0, \quad b_0 \rightarrow r_0.$$

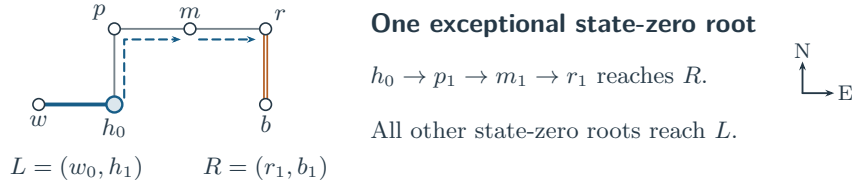


Figure 8: The exceptional path P_H , in order w, h, p, m, r, b with $h = (0, 0)$. The dashed route is the unique state-zero entry to R ; initially far pairs in different basins must reverse their spatial order before settling.

Thus every root again enters the left cycle.

For P_Z , write $z = (0, 0)$, $t = (0, 1)$, $n = (0, 2)$, $w = (-1, 1)$, $r = (1, 0)$, $b = (1, -1)$. The left cycle is (z_0, t_0) . The leaves n, w both enter t_1 , and the *NSW* rule sends that tag to z_0 . The other roots are on the cycle or satisfy

$$r_0 \rightarrow z_0, \quad b_0 \rightarrow r_0.$$

Consequently the east-going tag z_1 is unreachable from the distinguished initial state. $\square$

Lemma 5.22 (The exceptional path). *All pairs of state-0 roots on P_H achieve distance-one contact.*

Proof. Name its vertices

$$w = (-1, 0), \quad h = (0, 0), \quad p = (0, 1), \quad m = (1, 1), \quad r = (2, 1), \quad b = (2, 0).$$

Their induced path order is w, h, p, m, r, b . The complete twelve-tag map is

Cell	Mask	From 0	From 1
w	E	h_1	h_1
h	NW	p_1	w_0
p	ES	h_1	m_1
m	EW	p_0	r_1
r	SW	m_0	b_1
b	N	r_0	r_1

Its two tagged cycles are $L = (w_0, h_1)$ and $R = (r_1, b_1)$. The only state-0 root reaching R is h_0 , along $h_0 \rightarrow p_1 \rightarrow m_1 \rightarrow r_1$. Every other root reaches L : w_0 lies on it and p_0, m_0, r_0, b_0 move successively left towards it.

Pairs in the same basin contact by Lemma 5.12. For a cross-basin pair, one root is h . Its only predecessor in path order is w , which is initially adjacent. Every other root lies to the right of h , yet its eventual support L lies to the left of R . These walks reverse their strict order and therefore contact by Lemma 5.14. The argument includes all entry times and the complete transients. $\square$

Proposition 5.23 (Small subcubic trees). *Two copies of A^* , both started in state 0, achieve distance-one contact on every induced grid tree of maximum degree at most three with at most six cells.*

Proof. The one- and two-cell cases succeed at time zero. On a larger finite tree every trajectory eventually enters a tagged cycle, whose support is a path by Lemma 5.11. If a long support exists, its tagged cycle is unique by Lemma 5.19, so Lemma 5.12 applies. If all supports are edges, there is either one tagged cycle, handled by the same lemma, or the maze is one of the four configurations of Lemma 5.20. Lemma 5.21 and then Lemma 5.12 handle P_N, P_E, P_Z ; Lemma 5.22 handles P_H . $\square$

Trees containing a degree-four vertex are covered by the full-centre analysis below. The distinction matters: at such a vertex, two different tagged cycles may share a spatial endpoint in opposite states, a configuration excluded by Lemma 5.18 only under its stated degree bound.

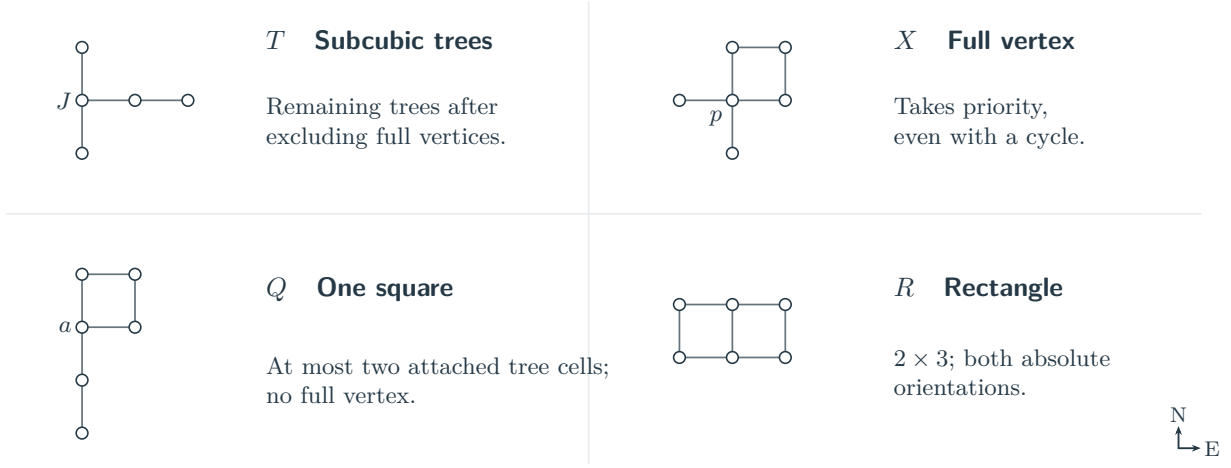


Figure 9: The structural partition for $3 \leq |V| \leq 6$, at a common grid scale. These are geometric families, not compass symmetry classes; no start pair is selected.

5.4 The remaining small mazes

We now account for full vertices and spatial cycles. This part of the proof works for arbitrary initial tags. Compass directions remain absolute throughout: translations are used to name cells, but no rotation of the fixed controller A^* is assumed.

Lemma 5.24 (Small-maze geometry). *Every connected induced grid maze with $3 \leq |V| \leq 6$ belongs to exactly one of the following structural families, with X given priority: X , a full vertex, its four neighbours, and at most one further cell; T , a tree of maximum degree at most three; Q , a unit square with at most two attached tree cells, with no full vertex and no second square; or R , either absolute orientation of the 2×3 rectangle.*

Proof. A degree-four vertex already accounts for five cells, so connectivity gives X . In its absence an acyclic maze belongs to T . A shortest spatial cycle in the remaining maze has length four or six, because the grid is bipartite. A simple six-edge grid cycle has two vertical and four horizontal edges, or four vertical and two horizontal edges. In the first case the two vertical edges join two adjacent rows, and each row contributes the straight two-edge segment between their endpoints. The six vertices form a 3×2 rectangle. Inducedness supplies its middle vertical edge, and hence a unit square. Interchanging the roles of the axes proves the same geometric assertion in the other case.

Thus a unit square is present and at most two cells remain. If these cells create another spatial cycle, they supply a three-edge detour between two adjacent square vertices. A one-cell exterior detour is impossible: the common grid neighbours of opposite square vertices are already in the square. The three-edge detour supplies a second unit square sharing an edge with the first, and their union is a 2×3 rectangle. Otherwise the remaining cells are trees attached at single square vertices, giving Q . The four families are disjoint under the stated priority. $\square$

Lemma 5.25 (Full-centre routing). *On every maze in X , two copies of A^* rendezvous from arbitrary initial tags.*

Proof. Translate the full vertex to $p = (0, 0)$, and write $n = (0, 1)$, $e = (1, 0)$, $s = (0, -1)$, $w = (-1, 0)$. If a sixth cell x is present, it must neighbour an arm, so

$$x \in \{(0, 2), (2, 0), (0, -2), (-2, 0), (1, 1), (-1, 1), (1, -1), (-1, -1)\}.$$

Table 8 lists all tagged cycles. A parenthesized word is read cyclically. We verify both the words and their exhaustion directly from Table 3.

In the first row, $p_0 \rightarrow e_0 \rightarrow p_1 \leftrightarrow s_1$, while $e_1 \rightarrow p_0$ and $s_0 \rightarrow p_0$. With no extra cell the north and west arms send both tags to p_1 . For a north extension both tags at n go to p_1 , and both tags at x

Extra cell	Complete list of tagged cycles	Support type
None, $(0, 2), (-2, 0), (-1, 1)$	(p_1, s_1)	One edge
$(0, -2)$	(s_1, x_1)	One edge
$(2, 0)$	$(p_0, e_0), (p_1, s_1)$	Edges meeting at p
$(1, 1)$	$(p_0, e_0, x_1, e_1), (p_1, s_1)$	Shuttle and edge at p
$(1, -1)$	(p_0, e_0)	One edge
$(-1, -1)$	(s_1, x_0, w_0, x_1)	Path $s - x - w$

Table 8: All full-centre cases through six cells, grouped by their tagged recurrence. Coordinates are absolute relative to the full vertex p .

go to n_1 . For a west extension, $w_0 \rightarrow x_0 \rightarrow w_1 \rightarrow p_1$, and $x_1 \rightarrow w_1$. For the northwest diagonal the remaining transitions are

$$w_0 \rightarrow x_0 \rightarrow w_1 \rightarrow p_1, \quad n_0 \rightarrow x_0, \quad n_1 \rightarrow p_1, \quad x_1 \rightarrow n_1.$$

These routes include every tag in those four geometries.

For a south extension both tags at s go to x_1 , and $x_1 \rightarrow s_1, x_0 \rightarrow s_0$. The north and west leaves enter p_1 ; the east leaf returns to p_1 from state zero and to p_0 from state one. Since $p_0 \rightarrow e_0 \rightarrow p_1 \rightarrow s_1$, every tag reaches the displayed outer edge.

For an east extension the horizontal straight rule gives $p_0 \leftrightarrow e_0$, and $p_1 \leftrightarrow s_1$. All other tags enter those words through

$$e_1 \rightarrow x_1 \rightarrow e_0, \quad x_0 \rightarrow e_1, \quad n_q \rightarrow p_1, \quad w_q \rightarrow p_1, \quad s_0 \rightarrow p_0.$$

Here and below $q \in \{0, 1\}$. For the northeast diagonal the east arm has mask NW, and x has mask SW, producing $p_0 \rightarrow e_0 \rightarrow x_1 \rightarrow e_1 \rightarrow p_0$. The southern edge remains (p_1, s_1) , and the remaining tags satisfy

$$n_0 \rightarrow p_1, \quad n_1 \rightarrow x_1, \quad w_q \rightarrow p_1, \quad x_0 \rightarrow n_0, \quad s_0 \rightarrow p_0.$$

In each of these two geometries one tagged cycle has edge support and the other visits p , so Lemma 5.13 applies to cross-cycle pairs.

For the southeast diagonal the east arm has mask SW, giving $p_0 \leftrightarrow e_0$. The other tags enter this edge through

$$p_1 \rightarrow s_1 \rightarrow x_1 \rightarrow s_0 \rightarrow p_0, \quad e_1 \rightarrow x_1, \quad x_0 \rightarrow e_1,$$

together with the north and west leaf returns to p_1 . Finally, for the southwest diagonal the masks of s, x, w are NW, NE, ES. They give $s_1 \rightarrow x_0 \rightarrow w_0 \rightarrow x_1 \rightarrow s_1$. The remaining tags follow

$$p_1 \rightarrow s_1, \quad p_0 \rightarrow e_0 \rightarrow p_1, \quad e_1 \rightarrow p_0, \quad n_q \rightarrow p_1, \quad w_1 \rightarrow p_1, \quad s_0 \rightarrow p_1.$$

Every single-cycle case has path support, and every tagged cycle in the two-cycle cases also has path support. Lemma 5.12 handles pairs entering the same tagged cycle. This proves the claim for all nine geometric possibilities and all initial tags. $\square$

Lemma 5.26 (Short twig returns). *Attach a one- or two-cell path to a core vertex p , with no further neighbours at the new cells, and stop a trajectory at its first return to p . For a one-cell twig in direction d , the returned state is one if $d \in \{N, W\}$, is the incoming state if $d = S$, and is its complement if $d = E$.*

For a two-cell twig $p - u - v$, write de for the directions from p to u and from u to v . All four twig tags return, except for the following three cases: ES has the internal edge cycle (u_1, v_1) , with u_0, v_0 returning to p_0 ; SS sends every tag to (u_1, v_1) ; and SW sends every tag other than u_0 to (u_1, v_0) , while $u_0 \rightarrow p_1$. Every return in the other cases takes at most four steps. The exact maps are given in Table 14.

Proof. The one-cell rule is the corresponding opposite-direction endpoint row of Table 3. For two cells there are exactly twelve words de , because $e \neq -d$. Table 14 prints all four successors for each word, so following its arrows proves every stated return or absorption. For example, the EE row has $v_0 \rightarrow u_1 \rightarrow v_1 \rightarrow u_0 \rightarrow p_0$; the SS row has $u_0, u_1 \rightarrow v_1 \rightarrow u_1$ and $v_0 \rightarrow u_0$; and the SW row has

$u_0 \rightarrow p_1$, $u_1 \rightarrow v_0 \rightarrow u_1$, and $v_1 \rightarrow u_1$. The appendix verifies the remaining rows and the square-port restrictions used below. $\square$

Lemma 5.27 (Square routing). *On every maze in $\mathbb{Q}$, two copies of A^* rendezvous from arbitrary initial tags.*

Proof. Translate the unique square so that

$$a = (0, 0), \quad b = (1, 0), \quad c = (1, 1), \quad d = (0, 1).$$

At most one exterior port occurs at each corner, because full vertices were assigned to X. Table 9 gives all corner responses. Expressions such as $(a + S)_1$ name the exterior neighbour and its arriving state.

Corner	Mask	Successor from state 0	Successor from state 1
a	NE	d_0	b_1
	NES	$(a + S)_1$	b_1
	NEW	$(a + W)_0$	b_1
b	NW	c_1	a_0
	NEW	a_0	$(b + E)_1$
	NSW	$(b + S)_0$	$(b + S)_0$
c	SW	d_0	b_1
	NSW	b_0	b_0
	ESW	b_1	b_1
d	ES	a_1	c_1
	NES	a_1	c_1
	ESW	a_1	a_1

Table 9: Every possible nonfull square-corner row. Exterior ports occur in at most two additional cells in total.

The exterior ports at c and d are never selected. This fact alone does not exclude an external attractor. At ports cN, cE, dN, dW , the allowed two-cell words are, respectively,

$$\{\text{NN}, \text{NE}\}, \quad \{\text{EE}, \text{EN}\}, \quad \{\text{NN}, \text{NW}\}, \quad \{\text{WW}, \text{WN}\}.$$

The omitted continuation would touch the neighbouring square corner and produce R. Lemma 5.26 proves that every tag in every allowed dormant twig returns to the square. Once such a tag has returned, the twig cannot be entered again. Also d can be entered recurrently only from a by a northward move arriving in state zero, or from c by a westward move arriving in state zero. Hence d_1 is transient, even though a dormant-twig root may visit it once.

The active ports are aS, aW, bE, bS . Their complete return rules are in Table 15. If one port already uses a cell, each other twig has at most one cell. Every tag on a one-cell twig returns immediately to the square; the specific return states used below refer to its entry from the core. The possible masks at b now give four routing regimes.

First, let $b = \text{NW}$ and $c \in \{\text{SW}, \text{ESW}\}$. Both tags at c feed the left core, whose fixed route is

$$d_0 \longrightarrow a_1 \longrightarrow b_1 \longrightarrow a_0.$$

If $a = \text{NE}$, it closes through $a_0 \rightarrow d_0$, giving the unique tagged cycle (d_0, a_1, b_1, a_0) , with path support $d - a - b$. If a has a south or west port, only a_0 leaves through it. A one-cell twig returns to a_1 , as does every tag on a two-cell west twig. The resulting unique tagged cycle is a shuttle extending $b - a$ by that twig. A two-cell south twig has word SS or SW , and a_0 enters it as u_1 , which belongs to its absorbing basin. In the latter case the only returning tag is u_0 , and it follows $u_0 \rightarrow a_1 \rightarrow b_1 \rightarrow a_0 \rightarrow u_1$. Thus all roots reach the same internal edge cycle. The unused tag b_0 enters c_1 , and all other exterior tags return from dormant twigs, so no other tagged cycle remains.

Second, let $b = \text{NW}$ and $c = \text{NSW}$. The north port at c costs at least one exterior cell. There is an edge cycle $b_0 \rightarrow c_1 \rightarrow b_0$. The left route $d_0 \rightarrow a_1 \rightarrow b_1 \rightarrow a_0$ closes through d_0 , or through a one-cell twig at a returning to a_1 . That twig cannot have two cells because of the port at c . Hence the other

tagged cycle is a four-tag shuttle on a three-vertex path and contains b_1 . The dormant twig at c returns and then enters b_0 ; a dormant twig at d also returns to the square. The displayed corner transitions exhaust all other tags. The two supports share b , and one is an edge, so Lemma 5.13 applies.

Third, let $b = \text{NEW}$. There is an eastern twig, $b_0 \rightarrow a_0$, and b_1 exits east in state one. Every square tag eventually reaches b_1 : the tags at c enter b_0, b_1 , or d_0 , while $d_0 \rightarrow a_1 \rightarrow b_1$. At a_0 , either the route through d_0 reaches b_1 , or a twig at a returns to a_1 and then b_1 . Such a twig has at most one cell. A one-cell eastern twig returns to b_0 , and a two-cell EE twig returns every tag to b_0 . In these cases there is one tagged cycle, supported on the path from the left cap (d , or the single twig cell at a) through a, b to the eastern twig. If the eastern twig has word ES, the departure from b_1 enters its edge absorber. Its returnable tags u_0, v_0 first return to b_0 , then reach b_1 , and next enter the same absorber. Thus this alternative has a unique edge-supported tagged cycle. Dormant twig returns again account for all exterior roots.

Fourth, let $b = \text{NSW}$. Both tags at b exit south in state zero. Every square tag reaches b ; any twig at a has at most one cell and returns to a_1 , which goes directly to b_1 . A one-cell southern twig returns to b_0 , giving an edge cycle. A two-cell SE twig also returns all tags to b_0 , and the tagged cycle still uses only b and its first twig cell. A two-cell SS twig captures every tag on its outer state-one edge. All other roots feed the indicated unique tagged cycle.

These regimes cover all possible corner masks. All displayed tagged cycles have path support. Lemma 5.12 handles same-cycle pairs, and Lemma 5.13 additionally handles the second regime. Hence all initial tags are covered. $\square$

Lemma 5.28 (The two rectangle orientations). *On both mazes in $\mathbf{R}$, two copies of A^* rendezvous from arbitrary initial tags.*

Proof. For the horizontal rectangle label the bottom row $a = (0, 0), b = (1, 0), c = (2, 0)$ and the top row $d = (0, 1), e = (1, 1), f = (2, 1)$. For the vertical rectangle use $a = (0, 0), b = (1, 0), c = (0, 1), d = (1, 1), e = (0, 2), f = (1, 2)$. The two complete maps are printed separately in Table 10.

Horizontal rectangle				Vertical rectangle			
Cell	Mask	From 0	From 1	Cell	Mask	From 0	From 1
a	NE	d_0	b_1	a	NE	c_0	b_1
b	NEW	a_0	c_1	b	NW	d_1	a_0
c	NW	f_1	b_0	c	NES	a_1	d_1
d	ES	a_1	e_1	d	NSW	b_0	b_0
e	ESW	b_1	b_1	e	ES	c_1	f_1
f	SW	e_0	c_1	f	SW	e_0	d_1

Table 10: All twelve tagged-state transitions in each absolute rectangle orientation. These are two separate controller calculations.

The horizontal map has the tagged cycle $(a_0, d_0, a_1, b_1, c_1, b_0)$, supported on $d - a - b - c$. Every remaining tag reaches it: $c_0 \rightarrow f_1 \rightarrow c_1$, $d_1 \rightarrow e_1 \rightarrow b_1$, $e_0 \rightarrow b_1$, and $f_0 \rightarrow e_0$. Lemma 5.12 proves contact.

The vertical map has the two tagged cycles (a_0, c_0, a_1, b_1) and (b_0, d_1) , supported on $c - a - b$ and $b - d$. The remaining tags satisfy $c_1, d_0 \rightarrow d_1, b_0$, respectively, $e_0 \rightarrow c_1$, $e_1 \rightarrow f_1 \rightarrow d_1$, and $f_0 \rightarrow e_0$. Thus these words exhaust recurrence. Lemma 5.12 handles same-cycle pairs, and Lemma 5.13 handles cross-cycle pairs at the common vertex b . $\square$

Proposition 5.29 (Six-cell success). *The controller A^* succeeds from every pair of state-zero starts on every connected induced grid maze with at most six cells.*

Proof. On one or two cells every pair is already successful at time zero. For larger mazes Lemma 5.24 is exhaustive. The family $\mathbf{T}$ is covered by Proposition 5.23; the other three families are covered by Lemmas 5.25, 5.27, and 5.28. $\square$

5.5 Sharpness and restricted adversaries

Theorem 5.30 (The attaining controller). *The controller A^* succeeds from every state-0 start pair on every legal induced grid maze with at most six cells. Moreover,*

$$\tau(A^*) = \tau_{\text{grid}}(A^*) = \tau_{\mathcal{J}}(A^*) = 7.$$

Proof. Proposition 5.29 proves success by combining the priority classification of Lemma 5.24, Proposition 5.23, and Lemmas 5.25, 5.27, and 5.28. Lemma 5.10 supplies a seven-cell path trap, which belongs to each displayed restricted class. There is no smaller trap even in the unrestricted class. $\square$

Proposition 5.9 gives the opposite bound for every controller with at most two states, with all its witnesses in $\mathcal{J}$. The attaining controller therefore proves Theorem 5.1. This deduction uses no auxiliary classification of controllers on paths.

The constant seven has a geometric explanation within the proof. The two vertical state-1 edge cycles of the U witness each have only a northern external port. The port comparisons in the tree proof exclude a connection of length at most three between two such separated supports. A connector of length four can have direction word NEES, using three new cells in addition to the four support cells. At six cells, the mixed profiles that fit either have a state-0 gate blocking one basin or produce the exceptional path P_H , where basin inversion forces contact before recurrence. Seven is therefore the first available size for this separated-shuttle mechanism.

This explanation does not classify all minimal traps. A spatially cyclic seven-cell maze can fail by a different mechanism. For example, on

$$\{(0, 1), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (3, 1)\}$$

the four tags

$$(1, 0)_0 \longrightarrow (1, 1)_0 \longrightarrow (2, 1)_0 \longrightarrow (2, 0)_1 \longrightarrow (1, 0)_0$$

form a recurrent tagged cycle. Starting at its two state-0 opposite corners $(1, 0)$ and $(2, 1)$ maintains Manhattan distance two. This also shows why the same-cycle contact lemma cannot be used without its path-support hypothesis.

6 Paths and the memory hierarchy

The full-grid theorem of Section 5 has a sharp path-only analogue, but its proof has a different status: the upper bound is analytic, while the matching lower bound is an exact finite certificate.

Theorem 6.1 (Two states on induced paths). *For radius-one rendezvous on finite induced grid paths,*

$$F_{\mathcal{P},1}(2) = 8.$$

The upper bound $F_{\mathcal{P},1}(2) \leq 8$ is analytic. The lower bound $F_{\mathcal{P},1}(2) > 7$ is established by exhaustive finite verification. The equality therefore has a computer-assisted proof.

6.1 An analytic eight-cell upper bound

The path-forcing part of the two-state upper proof in Section 5 implies the following W9 consequence: if a two-state controller succeeds on every induced path with at most seven cells, then after one simultaneous dihedral symmetry of the compass and grid its table contains

$$\delta(0, N) = (N, 1), \quad \delta(1, SW) = (S, 0). \tag{4}$$

These are two rows of the nine-entry pattern of Lemma 5.7.

Consider the cells, in endpoint order,

$$\begin{aligned} p_0 &= (1, 0), & p_1 &= (1, 1), & p_2 &= (0, 1), & p_3 &= (0, 2), \\ p_4 &= (0, 3), & p_5 &= (1, 3), & p_6 &= (2, 3), & p_7 &= (2, 2). \end{aligned} \tag{5}$$

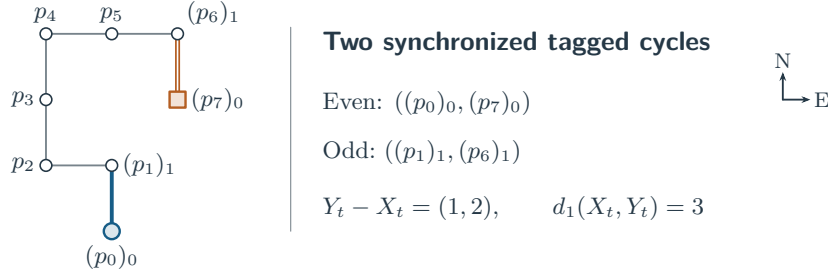


Figure 10: The eight-cell path obstruction under the two W9 rows in (4). The circle and square mark state-0 starts; both phases have the same displacement $(1, 2)$.

Their masks in this order are

$$N, \quad SW, \quad NE, \quad NS, \quad ES, \quad EW, \quad SW, \quad N.$$

There is no unit-distance pair among nonconsecutive cells, so the induced graph is exactly a path.

Start both agents in state 0 at p_0 and p_7 . The two rows in (4) give the complete executions

$$(p_0, 0) \longleftrightarrow (p_1, 1), \quad (p_7, 0) \longleftrightarrow (p_6, 1).$$

At even times the displacement is $p_7 - p_0 = (1, 2)$; at odd times it is $p_6 - p_1 = (1, 2)$. The Manhattan distance is therefore exactly three at every integer time.

If a controller with at most two states already has a path trap through seven cells, the desired upper bound is immediate. A one-state controller has a four-cell path trap by Corollary 4.2. Otherwise a genuine two-state path survivor satisfies the W9 consequence after simultaneous compass transport, and the inverse symmetry carries the displayed eight-cell trap back to the original controller. Hence

$$F_{\mathcal{P},1}(2) \leq 8.$$

6.2 Exact finite lower certificate

For the reverse inequality, use the two-state controller in Table 11.

Mask	state 0	state 1
N	(N, 1)	(N, 0)
E	(E, 1)	(E, 0)
S	(S, 0)	(S, 0)
W	(W, 0)	(W, 1)
NS	(S, 0)	(N, 1)
EW	(W, 0)	(E, 1)
NE	(E, 1)	(N, 1)
ES	(S, 0)	(E, 1)
NW	(W, 0)	(N, 1)
SW	(S, 0)	(S, 0)

Table 11: The two-state controller used in the exact finite lower certificate for paths. Rows of degree three and four may be completed arbitrarily because they are never queried on a path.

Proposition 6.2 (Finite path-survival certificate). *The controller of Table 11 succeeds from every state-zero start pair on every induced grid path with at most seven cells.*

Exact verification. The proof is computational and exhaustive. Paths are enumerated up to translation only, so absolute compass orientation is retained. The numbers of path shapes on $1, \dots, 7$ cells are

$$1, \quad 2, \quad 6, \quad 14, \quad 34, \quad 82, \quad 198,$$

for a total of 337. Across those paths there are exactly 4042 unordered start pairs whose initial Manhattan distance exceeds one. For each pair, the deterministic product trajectory is followed until

either distance-one contact occurs or a complete product tag repeats. A repeated product tag before contact is an exact infinite-failure certificate, so no time cutoff is used. All 4042 pairs succeed. An independently written connected-cell-set generator reproduces the same path sets through seven cells. The checker, data, and enumeration specification belong to the computational supplement.

Proposition 6.2 gives $F_{\mathcal{P},1}(2) > 7$. Together with the analytic upper bound it proves Theorem 6.1 with the stated mixed status.

6.3 Four states solve every finite path

The behavior changes qualitatively by four states. Let

$$Q = \{N, E, S, W\}$$

with any one label, say N, designated as the initial state. Fix the cyclic order

$$N, E, S, W.$$

For a nonempty mask M and state q , let $\text{succ}_M(q)$ be the first member of M encountered strictly after q in this cyclic order, wrapping around if necessary. Define

$$d = \text{succ}_M(q), \quad \delta(q, M) = (d, -d). \quad (6)$$

After the first move, the state records the compass port leading back to the cell just left; the next move takes the next open port in cyclic order.

Proposition 6.3 (Contour recurrence on grid trees). *On every finite induced grid tree T with at least two cells, the controller (6) has one recurrent cycle on the arrival-port tags*

$$H = \{(v, q) : q \in M_T(v)\}.$$

This cycle has length $2|E(T)|$ and traverses every directed tree edge exactly once. Every tag outside H enters H after one step.

Proof. If the controller moves from v in direction d , then at the new cell the state is $-d$, the port leading back to v . Hence every transition lands in H .

Restricted to H , the map takes an incoming port to the next incident port in the fixed cyclic order. It is a permutation: for an arrival tag (v, q) , the previous cell must be $u = v + q$; choosing at u the incident port immediately preceding $-q$ makes the next selected direction equal to $-q$, which lands at (v, q) .

To prove that this permutation has one cycle, induct on the number of edges. A single edge gives the two-tag shuttle. If z is a leaf adjacent to v , remove z . By induction the smaller tree has one contour cycle. Restoring the leaf inserts its incident port into the cyclic order at v : one old successor transition is replaced by the detour $v \rightarrow z \rightarrow v$ followed by the formerly chosen next edge. Thus the two directed tags of the restored edge are inserted into, rather than separated from, the existing cycle. Repeating this leaf insertion yields one cycle on all of H . Finally

$$|H| = \sum_{v \in V(T)} \deg_T(v) = 2|E(T)|.$$

□

This proposition is a contour statement, not a rendezvous theorem on branching trees. On a path, however, the contour support is itself a path, so Section 3 applies directly.

Theorem 6.4 (Universal four-state path controller). *The controller (6) succeeds from every start pair on every finite induced grid path at radius one.*

Proof. A singleton succeeds at time zero. On any nonsingleton path, Proposition 6.3 puts every trajectory on the same recurrent tagged cycle after at most one move. Its spatial support is the whole path. The two trajectories are therefore phase shifts of one periodic path walk, and Corollary 3.5 forces distance-one contact. □

Corollary 6.5 (Infinite path threshold from four states onward). *For every real $\rho \geq 1$ and every $s \geq 4$,*

$$F_{\mathcal{P},\rho}(s) = \infty.$$

Proof. The one fixed controller in Theorem 6.4 has no finite path trap at radius one, and hence none at any larger radius. It is admissible under every budget $s \geq 4$. Equation (2) gives the conclusion. $\square$

The remaining path transition is exactly the three-state case:

$$F_{\mathcal{P},1}(1) = 4, \quad F_{\mathcal{P},1}(2) = 8, \quad F_{\mathcal{P},1}(3) \text{ is undetermined}, \quad F_{\mathcal{P},1}(s) = \infty \ (s \geq 4).$$

The branching counterexample in Appendix D.5 explains why the two-state path-support mechanism does not extend automatically once three states are available; it does not settle the three-state path problem.

7 Three-state recurrent-cycle realization

A recurrent word must satisfy both capacity and shared-table compatibility. We characterize the latter on three-state paths; initial-state accessibility remains a separate question. Figure 11 previews the construction.

7.1 The normal form and its scope

Proposition 7.1 (Three-state path normal form). *Let a simple three-state tagged cycle have used path support with $n \geq 3$ vertices. Every edge multiplicity is one or two; the doubled edges form a matching. Up to cyclic shift its spatial word is the ordinary contour, with one passage of each doubled edge replaced by the three-step passage “forward, back, forward.” The choice of the forward or backward contour passage is its insertion phase.*

Proof. For adjacent edges, capacity gives $k_{j-1} + k_j \leq 3$ and both multiplicities are positive. Hence every k_j is one or two, and two doubled edges cannot be adjacent. A neighbouring single edge is crossed once in each direction. The additional two crossings of a doubled edge must therefore lie within one of its two contour passages; otherwise a neighbouring single-edge cut would be crossed again. They form the stated immediate backtrack. Endpoint insertions can have equivalent cyclic descriptions, so uniqueness of the parametrization is not asserted. $\square$

For a one-edge support the multiplicity can instead be three; that case is not a matching input and is treated separately in Lemma D.1. The normal-form necessity concerns used support and remains valid in a larger maze. The converse below uses the masks of the path itself. A proper interval of an ambient path requires the boundary-aware conditions in Appendix D.3.

Fix a finite induced grid path $P = (v_0, \dots, v_{n-1})$, $n \geq 2$, with its absolute compass orientation and exactly its own masks. Choose a matching $\mathcal{D}_2 \subseteq \{0, \dots, n-2\}$ and phases $\eta_e \in \{0, 1\}$. In the contour $0, 1, \dots, n-1, n-2, \dots, 1$, replace the index-increasing passage of e if $\eta_e = 0$, and its index-decreasing passage if $\eta_e = 1$, by the three-step backtrack. Denote this specification by S , and write

$$I = \mathbb{Z}/L\mathbb{Z}, \quad L = 2(n-1) + 2|\mathcal{D}_2|, \quad w_i \in V(P), \quad \sigma(i) = i+1, \quad M_i = M_P(w_i).$$

Let d_i be the direction from w_i to $w_{\sigma(i)}$. An occurrence means an index $i \in I$, not just its cell. Internal cells have load two or three, and endpoints load one or two. An internal cell of load three is *saturated*. Realization asks for a legal three-state table and states q_i such that the L distinct tags (w_i, q_i) follow this cyclic word. No primitive spatial-word hypothesis or state-zero accessibility requirement is imposed.

A necessary local condition is *same-mask doubled-side consistency*: all saturated cells of one absolute mask must use the same port twice. We call this the direction condition. The finite profile family and matrices in the next theorem are constructed explicitly below from masks, occurrences and successor links.



Repeated mask $M = NW$, paired port W :

$$Y_i + Y_j = \kappa_M(t_i + t_j)$$

$t_i = t_j$ forces the same successor.

$\kappa_M = 0$ allows merged outputs.

Figure 11: The realization construction, with $Y_i = X_{\sigma(i)}$. The same profile serves every repeated-mask component; varying the profile gives a finite union of affine solution sets, not one XOR system.

Theorem 7.2 (Exact three-state realization). *For every specification S just defined, the following are equivalent:*

- (i) S is realized by a simple three-state tagged cycle on the path with its own masks;
- (ii) the direction condition holds and some $\theta \in \Theta(S)$ satisfies

$$\text{rank}_{\mathbb{F}_2} A_\theta = \text{rank}_{\mathbb{F}_2} [A_\theta \mid b_\theta];$$

- (iii) the direction condition holds and there is one $\theta \in \Theta(S)$ such that

$$\forall \lambda : \quad \lambda^T A_\theta = 0 \implies \lambda^T b_\theta = 0.$$

From a consistent profile and its solution one constructs all 45 legal controller rows. The resulting cycle has least tagged period exactly $L = 2(n-1) + 2|\mathcal{D}_2|$. Every realizing state assignment belongs to at least one profile. The same profile is used throughout all components of the specification.

7.2 Two-input functionality and the finite profile

The algebraic mechanism concerns a function on two inputs, not an invertible successor map. Encode the three states by the nonzero vectors

$$H = \mathbb{F}_2^2, \quad Q_3 = H \setminus \{0\} = \{1, 2, 3\},$$

where the numbers are binary vector codes and addition is XOR. The final controller names are obtained by $1 \mapsto 0, 2 \mapsto 1, 3 \mapsto 2$.

Lemma 7.3 (Two-input successor difference). *For input bits $t_i \in \mathbb{F}_2$ and output states $Y_i \in Q_3$, the conditions*

$$t_i = t_j \implies Y_i = Y_j \quad \text{for all } i, j$$

are equivalent to the existence of $\kappa \in H$ with

$$Y_i + Y_j = \kappa(t_i + t_j) \quad \text{for all } i, j.$$

Proof. If only one input occurs, all outputs coincide and $\kappa = 0$ works. If both occur, functionality gives two outputs Y_0, Y_1 ; take $\kappa = Y_0 + Y_1$. Conversely, equal inputs make the right side zero and force equal outputs. The case $Y_0 = Y_1$ is allowed and is exactly the zero difference. $\square$

For two occurrences of $M = NW$ on its paired W -side, take $\kappa = 3 = (1, 1)$ and role bits $t_i = z_u, t_j = z_v$. If the observed successors are $Y_i = 1, Y_j = 2$, the lemma gives $3 = 3(z_u + z_v)$, hence $z_u + z_v = 1$ over $\mathbb{F}_2$. Equal outputs instead require equal bits for this nonzero difference; $\kappa = 0$ allows both inputs to share an output. This is an algebraic illustration, not a claim that a separately prescribed path is realizable.

Every internal vertex uses both ports. A realizing three-state direction map at its mask is therefore a surjection with fibres of sizes one and two. For each internal mask M choose its singleton port u_M , its singleton state $a_M \in Q_3$, and let p_M be the other port. If M has a saturated representative, p_M must be its doubled port, so u_M is forced; otherwise both port choices are retained. Choose a successor difference $\kappa_M \in H$, restricted to $\kappa_M \neq 0$ when M is saturated. For each endpoint separately choose an injective assignment of its actual occurrences to Q_3 . These choices form $\theta \in \Theta(S)$. They assign no arbitrary row outputs.

Let $b(a)$ be the smaller numeric code in $Q_3 \setminus \{a\}$. For every internal cell v introduce one bit z_v . Define its occurrence states by

$$X_i = \begin{cases} a_{M_i}, & d_i = u_{M_i}, \\ b(a_{M_i}) + a_{M_i}t_i, & d_i = p_{M_i}, \end{cases} \quad t_i = z_{w_i} + \epsilon_i. \quad (7)$$

At a load-two cell the unique paired-side occurrence has $\epsilon_i = 0$. At a saturated cell the two such occurrences have $\epsilon = 0, 1$ in temporal order after choosing an origin of the cyclic word. Endpoint X_i are the constants chosen in the profile. The two paired states are $b(a_M)$ and $b(a_M) + a_M$, both nonzero and different from a_M . Thus local injectivity is built in. The state absent over a load-two cell is still a possible row state of its mask; it cannot be discarded globally.

7.3 The canonical affine system

For an internal mask form

$$U_M = \{i : M_i = M, d_i = u_M\}, \quad B_M = \{i : M_i = M, d_i = p_M\}.$$

Both groups are nonempty. In each choose an anchor j , for example the least occurrence index. For every other occurrence i impose respectively

$$X_{\sigma(i)} + X_{\sigma(j)} = 0 \quad (i, j \in U_M), \quad (8)$$

$$X_{\sigma(i)} + X_{\sigma(j)} = \kappa_M(t_i + t_j) \quad (i, j \in B_M). \quad (9)$$

Endpoint occurrences are grouped by the *pair* consisting of their one-port mask and their chosen state. For every non-anchor member of such a group impose

$$X_{\sigma(i)} + X_{\sigma(j)} = 0. \quad (10)$$

In particular, equal endpoint masks cannot be ignored. Equal states at different masks do not identify rows, and different states at the same endpoint mask are not identified either.

For a fixed θ , all a_M , κ_M and endpoint assignments are constants. Expanding the vector equations into their two scalar coordinates gives

$$A_\theta z = b_\theta, \quad z \in \mathbb{F}_2^{n-2}. \quad (11)$$

Constant equations are retained: $0 = 1$ rejects a profile. For $n = 2$ there are no internal bits and the same construction is a zero-dimensional system.

A group of h occurrences contributes $h - 1$ vector comparisons. There are fewer than $2L$ scalar rows, each involving at most two source bits and two successor bits. The symbolic object consequently has size $O(n)$ and is constructed without solving realizability. If m_3 internal masks have saturated representatives and m_2 do not, then $m_3 + m_2 \leq 6$ and

$$|\Theta(S)| \leq 36 \cdot 9^{m_3} 24^{m_2} \leq 36 \cdot 24^6. \quad (12)$$

This deliberately coarse constant is independent of the path length. Profiles may overlap; uniqueness of representation is not required.

7.4 Proof of the realization theorem

Proof of Theorem 7.2. Necessity. Let a legal table realize the distinct tags. Encode its states in Q_3 . At every internal mask both directions occur, so its direction fibres have sizes one and two. At a

saturated cell all three states occur, with the two states on the doubled side distinct. Therefore every saturated representative of that mask doubles the same side. Extract u_M, a_M, p_M from the table. At each cell the paired state or ordered pair determines a bit z_v in (7). The endpoint states give the required injections.

All sources in U_M have the same state and mask; deterministic successor states give (8). In B_M , source-state equality is equivalent to equality of t_i . Lemma 7.3, with $Y_i = X_{\sigma(i)}$, gives a κ_M satisfying (9). When the mask is saturated, its two paired occurrences at one cell move to the same neighbour along the doubled edge. Their successor occurrences are distinct, and distinct tags over that neighbour have different states. Hence their difference is nonzero, as required in the profile. This also covers a doubled edge incident to an endpoint. Equal endpoint keys give exactly (10). Thus the extracted profile and bits solve (11); every row dependency has zero right-hand-side sum.

These comparisons exhaust repeated keys. Different masks mean different rows. For the same internal mask, singleton and paired states are disjoint, and each fibre is handled above. A one-port endpoint mask cannot equal a two-port internal mask. The remaining endpoint coincidences are precisely the groups in (10). In particular no additional successor-row condition is missing.

Linear consistency. Row elimination over $\mathbb{F}_2$ produces an inconsistent equation exactly when a sum of original rows has zero left side and right side one. If no such row occurs, assign the free bits and back-substitute the pivot bits. This proves the equivalence of rank consistency, the stated balance condition, and existence of a solution, including the zero-dimensional case.

Sufficiency and reconstruction. Take one consistent profile and a solution. Equations (7) produce nonzero state vectors. At a load-two cell the singleton and paired states are different. At a saturated cell the opposite ϵ values produce both paired states, distinct from each other and from the singleton. Endpoint injections are part of the profile. Hence all L resulting tags are distinct.

Apply the one global renaming to actual controller states $q_i \in \{0, 1, 2\}$ and prescribe every used row by

$$\delta(q_i, M_i) = (d_i, q_{\sigma(i)}). \quad (13)$$

Suppose two prescriptions have the same key. At an internal mask both sources lie in the same direction fibre. In the singleton fibre, adding their anchor equations gives equal successors. In the paired fibre, equal states give $t_i = t_j$, and adding their anchor equations yields

$$X_{\sigma(i)} + X_{\sigma(j)} = \kappa_M(t_i + t_j) = 0.$$

The direction already agrees. At an endpoint the direction is unique, and (10) gives equal successors for equal keys. Thus (13) is a function, not a conflicting set of assignments.

Every used direction belongs to its mask. For each still unassigned key (q, M) , $M \neq \emptyset$, choose any $d \in M$ and any next state. This produces all $3(2^4 - 1) = 45$ legal rows without changing a used row. If only one paired input was observed, its profile difference need not prescribe the absent row's output. In particular, one must not extrapolate it to a zero vector and treat that vector as a state. The observed difference may instead be replaced by zero without changing the realized assignments.

Starting at (w_0, q_0) , the prescribed row takes the controller to $(w_{\sigma(0)}, q_{\sigma(0)})$; induction follows the entire specified word. In each selected compass direction the next cell is the prescribed neighbour. The orbit returns after L steps, and the preceding L tags are distinct. Its least tagged period is therefore exactly L . This argument neither assumes nor needs that the spatial projection have least period L . $\square$

The same key comparisons close all successor equalities. Within a fixed profile, linear elimination tests or projects partial assignments; across profiles, one takes a union. Unsaturated vertices therefore have no automatic completion rule.

7.5 Why one XOR system is not enough

Consider the fixed specification

$$\text{directions ENENNNE}, \quad \mathcal{D}_2 = \{0, 2, 6\}, \quad \eta_e = 0,$$

whose masks are E, NW, ES, NW, NS, NS, ES, W. Its cyclic word is

$$(0, 1, 0, 1, 2, 3, 2, 3, 4, 5, 6, 7, 6, 7, 6, 5, 4, 3, 2, 1).$$

Let x denote the same/swap correspondence of the ordered west-state pairs (q_1, q_{19}) and (q_5, q_{17}) at the two saturated NW cells. Let y denote that of the east-state pairs (q_4, q_6) and (q_{10}, q_{12}) at the two saturated ES cells; zero means same and one swap. The two intervening NS cells are unsaturated.

Proposition 7.4 (A non-affine saturated correspondence relation). *For this fixed spatial specification, the exact extendible relation is*

$$\text{Ext}(x, y) = \{00, 01, 10\}.$$

Proof. The singleton NW/N rows give $q_3 = q_7$ and $q_4 = q_8$. The singleton ES/S rows give $q_{18} = q_{14}$ and $q_{19} = q_{15}$. If $x = 1$, then $q_{17} = q_1 \neq q_{19}$. Equality of the two NS/S source states, $q_{16} = q_{15}$, would force $q_{17} = q_{16} = q_{19}$ by their successor rows. Thus two different south states are needed. If $y = 1$, then $q_{10} = q_6 \neq q_4 = q_8$. Equality of the north states $q_8 = q_9$ would force $q_9 = q_{10}$, again a contradiction. Hence two different north states are needed. North and south states at the same mask cannot coincide. The assignment 11 would therefore need four states. Appendix C.4 gives full realizing state assignments for the other three choices. $\square$

An XOR solution set and its projections are affine, with one, two or four possible assignments on two bits, never three. Even auxiliary XOR variables therefore cannot describe these actual correspondences by one system. This excludes faithful correspondence encodings, not arbitrary hypothetical decision-only graphs. The insertion phases η_e are fixed, distinct from x, y ; the unpinned specification is realizable.

7.6 An infinite phase-locking family

Theorem 7.5 (Phase locking across arbitrary gaps). *Let $r \geq 1$ and $\ell_0, \dots, \ell_r \geq 1$. On the induced monotone path*

$$E^{\ell_0} N E^{\ell_1} N \dots N E^{\ell_r},$$

double all r north edges and no others, assigning phases $\eta_1, \dots, \eta_r$. The specification is realizable with three states if and only if all phases are equal.

Proof. The lower endpoint of each doubled north edge has mask NW and singleton west state w ; the upper endpoint has mask ES and singleton east state e . At the j th edge denote its temporal north states by A_j, B_j and its south states by C_j, D_j . Then $\{A_j, B_j, w\} = Q$ and $\{C_j, D_j, e\} = Q$. For either insertion phase the south rows give

$$C_j \mapsto B_j, \quad D_j \mapsto w.$$

Two pairs (C_j, D_j) and (C_k, D_k) either agree or are swapped. A swap would identify B_j with w , so the pairs agree; then $B_j = B_k$ and $A_j = A_k$. The north rows are

$$\begin{array}{c|cc} \eta_j & A_j & B_j \\ \hline 0 & C_j & e \\ 1 & e & D_j \end{array}$$

and distinct phases would identify a paired state with the singleton e . Thus all phases are equal, independently of the horizontal gaps.

For sufficiency, use the following rows; the two rows for each phase list the three outputs in state order 0, 1, 2:

(η, M)	0	1	2
(0, NW)	(W, 2)	(N, 1)	(N, 0)
(0, ES)	(E, 1)	(S, 2)	(S, 0)
(1, NW)	(W, 2)	(N, 0)	(N, 1)
(1, ES)	(E, 1)	(S, 0)	(S, 2)

For either phase add $\delta(1, \text{EW}) = (E, 1)$, $\delta(2, \text{EW}) = (W, 2)$, $\delta(2, \text{E}) = (E, 1)$, and $\delta(1, \text{W}) = (W, 2)$. Start at the left endpoint in state 2. At phase zero the forward doubled-edge sequence is $\text{NW}_1, \text{ES}_1, \text{NW}_2, \text{ES}_0$,

with backward sequence ES_2, NW_0 . At phase one the forward sequence is NW_1, ES_0 , and the backward sequence is ES_2, NW_2, ES_1, NW_0 . Here a mask with a subscript denotes the occurrence at the corresponding endpoint of that edge with that state. Horizontal segments preserve state 1 eastward and state 2 westward. The endpoint rows close the word; all occurrences at one cell have different states. Completing unused rows gives the required simple cycle. $\square$

With $r = 2$ and all gaps one, unequal phases give the six-vertex staircase. Its contradiction is equivalently the three constraints $x = 1$, $x + y = 0$, $y = 0$ from the general successor comparisons, whose sum is $0 = 1$; an explicit derivation is recorded in Appendix C.5. Minimality of six among residual obstructions is a finite-check statement, not needed for this proof. More generally, for any fixed path radius R , choose gaps $(1, h, 1)$ with $h > 2R + 2$ and phases $(0, 1)$. Every radius- R neighbourhood occurs in a realizable uniform-phase specification, but the whole specification is unrealizable. Thus bounded geometric neighbourhoods do not replace the global mask constraints. This is not a lower bound on the size of an algebraic certificate.

7.7 Recognition, reconstruction and the accessibility boundary

Enumerate the finite profiles after checking the direction condition, compile (11), and solve by Gaussian elimination. A successful branch constructs the table by (13); rejection requires failure of every admissible profile. With K bounded as in (12), this gives a coarse $O(Kn^3)$ time bound and $O(n^2)$ working space. Fixed-alphabet polynomiality is not asserted as a separate novelty: finitely many full tables already exist independently of n . The content is the special sparse normal form and constructive removal of arbitrary row outputs.

Realization does not determine state-zero accessibility: two completions on one four-vertex path preserve the same contour while making it accessible from every state-zero start or from none. Appendix D.4 gives both tables and a conditional uniqueness criterion. Universal three-state rendezvous remains open.

8 State-complexity consequences

Theorem 2.4 converts trap thresholds into memory requirements:

$$F_{C,\rho}(s) > n \iff M_{C,\rho}(n) \leq s.$$

The strict inequality on the left is essential.

8.1 Consequences of the exact low-state thresholds

For all induced grid mazes at radius one, Section 4 gives $F(1) = 4$, while the central theorem of Section 5 gives $F(2) = 7$. Therefore

$$\boxed{M(1) = M(2) = M(3) = 1}, \quad \boxed{M(4) = M(5) = M(6) = 2}, \quad (14)$$

and

$$\boxed{3 \leq M(7) < \infty}. \quad (15)$$

Indeed, $F(1) > n$ exactly for $n \leq 3$ among these horizons, and $F(2) > n$ exactly for $n \leq 6$. The strict lower bound at $n = 7$ follows because $F(2) = 7$ is not greater than seven; finiteness follows from Theorem 8.1 below.

For induced paths, Theorem 6.1 and Corollary 6.5 give the sharper hierarchy

$$M_{\mathcal{P},1}(n) = 1 \quad (n \leq 3), \quad M_{\mathcal{P},1}(n) = 2 \quad (4 \leq n \leq 7), \quad (16)$$

and

$$\boxed{M_{\mathcal{P},1}(n) \in \{3, 4\} \quad (n \geq 8)}. \quad (17)$$

The first equality is deductive; the exact value 2 on $4 \leq n \leq 7$ inherits the finite lower certificate of Theorem 6.1. The two-value alternative (17) is deductive: the analytic bound $F_{\mathcal{P},1}(2) \leq 8$ gives $M > 2$, and the four-state universal controller gives $M \leq 4$.

Size horizon	all induced grid mazes, $M(n)$	induced paths, $M_{\mathcal{P},1}(n)$
$1 \leq n \leq 3$	1	1
$4 \leq n \leq 6$	2	2
$n = 7$	$3 \leq M(7) < \infty$	2
$n \geq 8$	$3 \leq M(n) < \infty$; exact value open	3 or 4

Table 12: State-complexity consequences of the exact obstruction thresholds and the finite-horizon existence theorem. The last path entry is an exact two-value alternative, not an estimate.

The unresolved three-state path problem has an exact dual interpretation. If a three-state controller succeeds on every finite induced path, then $M_{\mathcal{P},1}(n) = 3$ for every $n \geq 8$. If no such controller exists, then the finite set of three-state tables has a finite maximum first-trap size $F_{\mathcal{P},1}(3)$, and Theorem 2.4 implies $M_{\mathcal{P},1}(n) = 4$ for all sufficiently large n . Thus deciding universal three-state path rendezvous is equivalent to deciding whether the eventual path state complexity is three or four.

8.2 Finite state complexity at every fixed horizon

The upper bound in (15) is a general finite-horizon fact. It must not be confused with a single finite controller that solves mazes of all sizes.

Theorem 8.1 (Finite-horizon solvability). *For every fixed finite n and every $\rho \geq 1$,*

$$M_{\mathcal{G},\rho}(n) < \infty.$$

The controller may depend on n but not on the maze or the starting cell.

Proof. Fix n . Store a bounded exploration record: visited coordinates relative to the start, their observed masks, the current relative coordinate, a DFS stack and a mode flag. Coordinates are updated internally from chosen compass moves, not read from the environment. Since every cell is within graph distance $n - 1$ of the start, they lie in $[-(n - 1), n - 1]^2$. The finite collection of possible records is a finite state set.

At each cell, move to the first unvisited neighbour in a fixed compass order and push the current coordinate onto the stack; when none remains, traverse the stack edge to the parent. Read a newly reached cell's mask on the transition that chooses its next move. Thus DFS needs no stationary processing step and eventually returns to the root with the complete relative map.

Choose the lexicographically largest mapped cell and its first open neighbour in fixed compass order. Maps from different starts differ by translation, which preserves both choices; both agents therefore select the same physical edge. Follow a stored route to its selected endpoint, then shuttle on that edge forever. The mode change takes the first route edge, or the selected edge itself when already at the endpoint, so every transition moves. Once both agents shuttle, their Manhattan distance is zero or one, proving success for every $\rho \geq 1$.

A singleton succeeds at time zero. Complete impossible-record rows by arbitrary legal moves. The resulting table depends on n , not on the maze or either start. $\square$

The radius restriction in Theorem 8.1 is genuine for this model: when $0 \leq \rho < 1$, opposite starts on a two-cell edge swap forever under compulsory synchronous motion, independently of the number of states.

The next two sections fix first a complete test menu and then one host for all two-state controllers, strengthening the obstruction quantifiers in the same model.

9 Finite obstruction bases

The two-state case proof supplies three levels of obstruction uniformity:

$$\begin{aligned}
&\text{controller-dependent trap complexity: } \forall A \exists I_A, \\
&\text{finite complete test family: } \exists \mathcal{U} \text{ finite } \forall A \exists I \in \mathcal{U}, \\
&\text{one universal host maze: } \exists U \forall A \exists (x_A, y_A) \in U^2.
\end{aligned} \tag{18}$$

Here A ranges over legal controllers with at most two states, and each test or start pair is initially noncontacting. The successive lines fix the controller, then a menu, then the host itself. Finite controller space already gives an abstract finite menu from $F(2) < \infty$; the construction below instead extracts a small explicit family from the proof without enumerating controllers.

9.1 An explicit proof-derived obstruction basis

A *test instance* is a triple $I = (V, x_0, y_0)$ consisting of an absolutely oriented finite connected induced grid maze and two distinct state-0 starts with $\|x_0 - y_0\|_1 \geq 2$. We say that I *traps* A when the two synchronous copies of A started at x_0, y_0 never reach Manhattan distance at most one. Compass orientation is part of the instance.

Theorem 9.1 (Explicit finite obstruction basis). *There is an explicit family $\mathcal{U}_{\text{hum}}$ of 144 absolutely oriented, start-labelled test instances, each on at most seven cells, such that every legal deterministic controller with at most two states is trapped by at least one member of $\mathcal{U}_{\text{hum}}$.*

The family uses 76 fixed oriented maze geometries. Under simultaneous D_4 transport and exchange of the two identical agents it has 22 instance orbits, and its maze geometries have 13 D_4 -types.

Proof. The upper proof has 30 primitive terminal predicates. Making explicit the two equal-direction variants and the four reset-bit variants produces 34 normalized source records. In every branch the proof applies one global element $g \in D_4$ to the controller, masks, directions, maze and starts; it never rotates only the maze. The selected normalized trap is therefore transported back by g^{-1} . We take all eight inverse transports of all 34 source records, then quotient only by common translation, exchange of the two identical state-0 agents, and literal identity. Rotation and reflection are *not* quotient operations on the concrete family, because the compass is absolute. This yields exactly the 144 instances stated above.

The completeness implication itself is the original case tree. For an arbitrary two-state controller that tree reaches one of its terminal source records; the corresponding normalized execution is trapped by the terminal lemma. Simultaneous D_4 conjugacy preserves tagged executions and Manhattan distance, so the inverse transport traps the original controller. That transported instance belongs to $\mathcal{U}_{\text{hum}}$. A one-state controller is included by viewing it as a two-state table whose state-0 rows always return state 0. $\square$

The counting in Theorem 9.1 is finite bookkeeping rather than a premise of completeness. The 34 normalized source records and all 144 transported instances are supplied in the supplementary data. Their terminal conditions organize into the eight proof modules summarized in Table 13.

9.2 A computer-minimized subfamily

The proof-derived family is intentionally faithful to the proof rather than optimized for cardinality. Exact computation gives a smaller complete menu, but its scope must be stated carefully.

Proposition 9.2 (Minimum within the proof-derived candidate family). *There exists a complete test family of 44 instances, all chosen from $\mathcal{U}_{\text{hum}}$. Moreover, no complete subfamily of $\mathcal{U}_{\text{hum}}$ has fewer than 44 members.*

The existence and candidate-restricted optimum in this proposition are computer-assisted results, unlike the deductive completeness of the full family.

Module	Regime / first failed condition	Geometry and noncontact mechanism
AX	Straight directions coincide, or a normalized opposed-direction toggle/reset/endpoint test fails.	Line or U-path, ≤ 7 cells. Terminal shuttles or disjoint line supports; parity covers adjacent supports.
RP	After the front tests, both axes reset: $C_i, C_j \in \{C_0, C_1\}$.	Two three-edge arms, 7 cells. Closed pockets leave the shared corner unvisited.
Z	One axis is C_0 , the other I ; exchange axes if needed. Six successive corner alternatives.	L/U/offset path, ≤ 7 cells. Closed arm regions or forced prefixes; separation or parity.
O	One axis is C_1 , the other I . Seven successive corner alternatives.	L/U/offset path, ≤ 7 cells. Closed pockets and forced corner prefixes; separation or parity.
LOCK	Both axes are I . After drift/diagonal normalization, the first of three corner-direction tests fails.	Paired terminal-arm path, ≤ 7 cells. Locked disjoint translated supports.
BIT	All three direction tests pass; the first of four required corner output bits fails.	Six-cell path. Closed tagged regions or a prefix into disjoint cycles; gap or parity.
TAIL	Lock and bit tests pass, but the final two W9 entries do not both hold.	Seven-cell path. A forced shuttle versus a separated left region, including the transient.
J	W9 holds. For $u = \delta(0, NSW)$, $v = \delta(1, NSW)$, choose G2 if $v = (S, 0)$ or $[u = (S, 0), v \in \{(N, 0), (W, 1)\}]$; otherwise G1.	Seven-cell one-junction tree. Exact tagged words separate the executions through the transient.

Table 13: The eight terminal modules: C_0, C_1 are reset next-state maps and I is identity. These selectors guide the complete terminal proofs; only J needs a junction rather than a path.

Certificate sketch. For the upper bound, the full physical transition function of each of the 144 instances was reconstructed, and a SAT encoding of a legal controller escaping the selected 44 instances was proved unsatisfiable by an independently checked LRAT certificate. For the lower bound, there are 44 explicit legal controllers $P_1, \dots, P_{44}$. If

$$S_i = \{I \in \mathcal{U}_{\text{hum}} : I \text{ traps } P_i\},$$

then each S_i is nonempty and the 44 sets are pairwise disjoint. Any complete subfamily of the 144 candidates must therefore choose a distinct member from each S_i . This packing argument is a direct exact finite check of the full physical instances and does not use SAT.

The 44 tests are complete, but optimality is *restricted to the 144 supplied candidates*. Whether an arbitrary complete family can have at most 43 members remains open.

10 A universal host maze

Joining traps can change a queried compass mask even when the new edge was never traversed. A universal induced maze therefore requires protection of the entire execution, not just its recurrent edges.

A five-cell path already shows the issue. Number a vertical path 0, 1, 2, 3, 4 from bottom to top and start the agents at 0 and 4. There are legal completions with straight-axis rows for which the complete spatial words are

$$(0, 1, 2, 1)^\infty, \quad (4, 3)^\infty.$$

The distances are 4, 2, 2, 2 and then repeat, but every old cell is visited. Any proper connected induced extension changes the mask of some visited old cell. Thus one cannot demand that each original trap already contain a clean attachment site.

10.1 Clean attachment

Lemma 10.1 (Clean attachment). *Let V be an induced grid maze and let*

$$(x_t, q_t, y_t, r_t)_{t \geq 0}$$

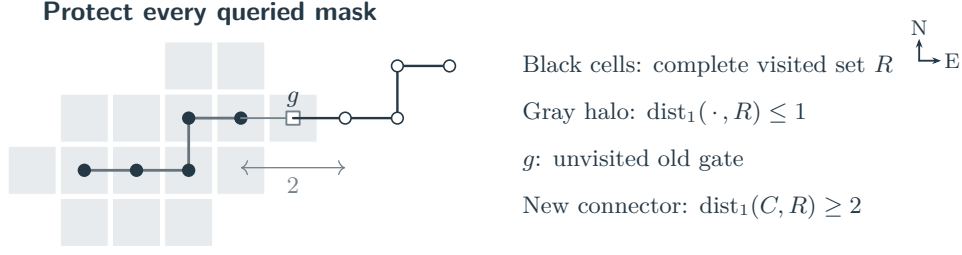


Figure 12: Clean attachment: the gray distance-one halo protects every cell in the complete visited set R . The new connector meets only an unvisited old gate and stays outside the halo; the drawing illustrates the clearance condition, not a particular trap.

be a complete trapped execution in V . Put

$$R = \{x_t : t \geq 0\} \cup \{y_t : t \geq 0\}.$$

Let C be a finite set of new cells, disjoint from V , such that the induced extension $V' = V \cup C$ is a maze and

$$\text{dist}_1(C, R) \geq 2.$$

Then the tagged execution from the same starts is identical in V and V' . In particular the extension introduces no rendezvous.

Proof. Induct on time. Suppose that both tagged positions agree through time t . Each current cell belongs to R , and no new cell is a grid neighbor of it. Its complete compass mask is therefore unchanged. The controller queries the same row, returns the same direction and next state, and the selected old neighbor is still present. Hence both next tagged positions agree. The conclusion follows simultaneously for the two agents. $\square$

The set R must include transient visits. New edges at unvisited old cells are harmless: the lemma protects queried masks, not every old mask.

10.2 Unused-cut stretching

The five-cell example above still has an unused edge cut. Stretching such a cut makes room for a protected port without modifying the masks of the old cells.

Lemma 10.2 (Unused-cut stretching). *Let the vertical cut $x = k + \frac{1}{2}$ cross at least one edge of an induced grid maze V , and suppose that neither complete trapped trajectory ever traverses an edge that crosses this cut. Define*

$$\phi(x, y) = \begin{cases} (x, y), & x \leq k, \\ (x + 4, y), & x \geq k + 1. \end{cases}$$

For every old crossing edge $(k, y)(k + 1, y)$ insert the four cells

$$(k + 1, y), (k + 2, y), (k + 3, y), (k + 4, y)$$

between its image endpoints (k, y) and $(k + 5, y)$. Then every old image cell has exactly its original compass mask, every actually selected move of the trapped execution joins the images of the same two old cells, and Manhattan distances between old image cells do not decrease. Moreover every inserted cell in the middle column $x = k + 2$ has Manhattan distance at least two from every old image cell.

Proof. Within either half-plane, ϕ is a translation, so noncrossing adjacencies and compass directions are unchanged. At a crossing edge the left endpoint still has an east neighbor and the right endpoint still has a west neighbor, now through the inserted straight corridor; no inserted cell is adjacent to any other old image. Thus every old mask is exactly preserved.

Mask preservation would not by itself preserve an execution that actually used a crossing edge, because such a move would enter a new corridor cell. The unused-cut hypothesis excludes this: every

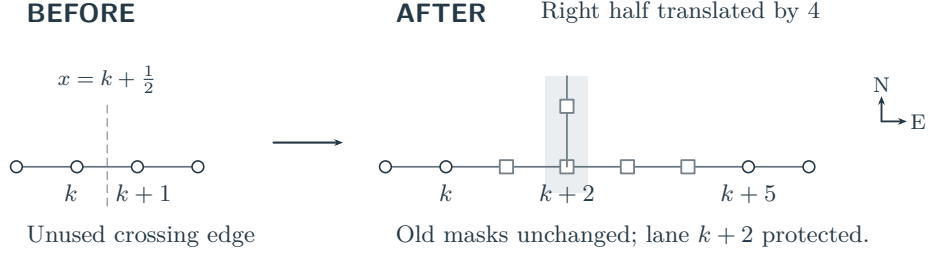


Figure 13: An unused crossing edge before and after width-four stretching. Four inserted cells preserve the old endpoint masks; the lane $x = k + 2$ has clearance for a protected branch.

selected old edge lies wholly in one half-plane and therefore maps to the same old adjacency. Induction on time transports the entire tagged execution.

Distances on one side are unchanged. For points on opposite sides the horizontal separation increases by four, so Manhattan distance cannot decrease. Finally, old left images have $x \leq k$ and old right images have $x \geq k + 5$; a cell with $x = k + 2$ is therefore at horizontal distance at least 2 from the former and at least 3 from the latter. This is the protected lane required for Lemma 10.1. $\square$

The horizontal version is obtained by writing the same geometric construction with x and y interchanged. This is a statement about the embedding of a fixed execution; it does *not* rotate or reflect the fixed controller.

For the cuts used below exactly one old tree edge crosses the cut. The replacement therefore subdivides that edge, and a simple route from the protected middle lane to an external port preserves the tree property and maximum degree three.

10.3 Portability of the junction leaves

All nonjunction leaves of the case tree are paths and admit an explicitly proved unused cut through an edge not traversed by either complete execution. The two final junction leaves require separate arguments.

For G2, the W9 symbolic words confine one execution to the cells B, J, A, U while the other alternates on its separated two-cycle; the boundary cell V is never visited. Its outward ray is therefore a clean port. For G1, one execution is the fixed word

$$B^0, D^1, F^1, D^1, F^1, \dots$$

and neither visits V nor traverses JB . The other begins A^0, J^1 . The G1 proof implies the dichotomy

$$V \text{ is never visited} \quad \text{or} \quad JB \text{ is never traversed by either execution.}$$

Indeed, a first visit to B can occur only as B^1 and immediately repeats the already visited tagged state J^1 , while a visit to V enters the permanent U^1, V^1 two-cycle. The same execution therefore cannot visit both B and V . In the first alternative we use the clean outward ray of V ; in the second we stretch the unused cut through JB . Both G1 variants are included in the fixed host in advance. The choice of which one supplies the witness may depend on the controller.

10.4 Assembly of the universal tree

Applying these port constructions to the inverse compass transports required by the case tree gives 172 ported fragments: 156 unused-cut fragments and 16 clean-boundary fragments. The number exceeds 144 because one oriented test can have more than one relevant proof provenance or attachment method, and G1 requires both of its portable alternatives in advance.

Place the 172 fragment boxes from left to right with two empty columns between successive boxes. Extend each exposed port by a vertical stalk to one top horizontal rail, and join the stalks along that rail. The separation prevents unintended induced adjacencies between fragments or stalks, while each fragment has exactly one connection to the global rail. The local pieces are trees of maximum degree

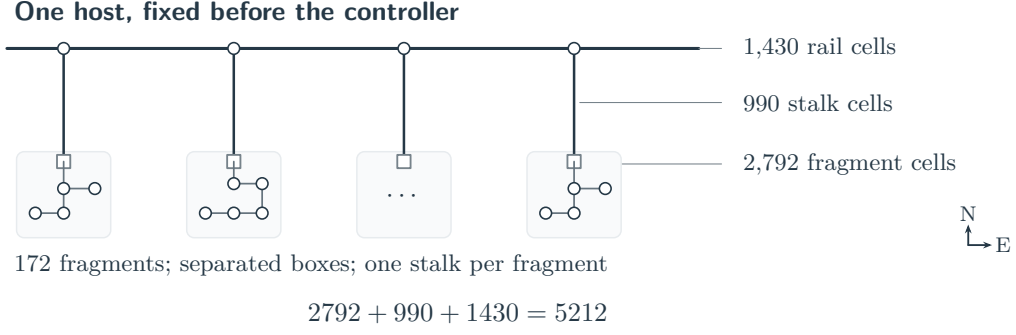


Figure 14: Universal-host assembly, shown schematically. Each separated fragment joins the rail through one stalk; the three displayed cell counts are disjoint contributions to the exact construction.

three, the stalks are paths, and the rail joins them without cycles. The induced union is therefore a subcubic tree.

The exact audited realization contains 2,792 cells in the disjoint ported fragments, 990 stalk cells strictly between the fragment ports and the rail, and 1,430 rail cells. Hence

$$|U| = 2792 + 990 + 1430 = 5212. \quad (19)$$

Its induced graph has 5,211 edges and maximum degree 3.

Theorem 10.3 (One universal two-state maze). *There exists a fixed finite connected induced grid tree U with 5,212 cells and maximum degree three such that for every legal deterministic controller A with at most two states there are state-0 starts $x_A, y_A \in U$, with $\|x_A - y_A\|_1 \geq 2$, for which the two copies of A never reach Manhattan distance at most one. Equivalently, the quantifier order is*

$$\boxed{\exists U \forall A \exists (x_A, y_A).}$$

Proof. Fix A . The case tree and Theorem 9.1 choose a terminal source trap together with one simultaneous compass normalization. Its inverse-transported oriented instance is represented among the fixed ported fragments. For a path leaf, its proved full-execution unused cut gives the protected embedding; for G2 use the unvisited gate; for G1 use the clean- V /unused- JB dichotomy. The local attachment lemmas preserve the complete unsuccessful execution. The separated global placement adds no new neighbor to any cell visited by that execution, so it is preserved once more in the same fixed maze U . Only the choice of fragment and its recorded start pair depends on A . $\square$

The obstruction basis and portability lemmas prove existence constructively. The exact 5,212-cell placement is separately validated from all 172 fragment records, their starts and embeddings, and the final induced graph; coordinate verification does not replace the assembly proof.

10.5 The universal-maze parameter

Theorem 10.3 suggests a parameter different from the controller-dependent quantity $F(2)$. Let

$$u_2 = \min\{|U| : U \text{ is a finite connected induced grid maze and} \\ \forall A, |Q_A| \leq 2, \exists x_A, y_A \in U \text{ trapping } A\},$$

where the starts are state-0 and initially noncontacting. Theorem 10.3 shows that this minimum is finite and gives $u_2 \leq 5212$.

Corollary 10.4 (Every exact size above the construction). *For every integer $n \geq 5212$ there exists a universal induced grid tree of maximum degree three with exactly n cells.*

Proof. Extend the exposed right end of the top rail by an induced horizontal ray of exactly $n - 5212$ new cells. In the explicit placement the first new cell is at Manhattan distance 8 from the union of all copied original gadget-cell images, a superset of every preserved witness trajectory; distances only increase farther along the ray. Thus no mask queried by any preserved witness changes. The first new

cell has only the old rail endpoint as an old neighbor, so the extension is a path attached at one vertex. Connectedness, acyclicity, inducedness, and maximum degree three are preserved. $\square$

The opposite direction is currently computational.

Proposition 10.5 (Computer-assisted lower bound). *The universal-maze parameter satisfies*

$$u_2 \geq 8.$$

Consequently

$$\boxed{8 \leq u_2 \leq 5212.}$$

Verification basis. The lower bound is a quantifier-reversal statement. It does not follow from the fact that every individual two-state controller has some trap on at most seven cells. Instead, an exhaustive checker enumerates every connected fixed induced grid cell set through size seven and tests the eight compass conjugates of the extremal controller A^* . The controller A^* has no trap below seven cells. Among all seven-cell geometries, a single fixed geometry traps at most four of the eight conjugates, even when its start pair may be chosen separately for each conjugate. Thus no maze with at most seven cells is universal for this eight-controller subset. This proposition is computer-assisted; no deductive classification of all seven-cell universal candidates is claimed.

It is also useful to isolate the tree-restricted version

$$u_{2,\text{tree}3} = \min\{|U| : U \text{ is universal and is an induced grid tree with } \Delta(U) \leq 3\}.$$

The construction proves

$$8 \leq u_2 \leq u_{2,\text{tree}3} \leq 5212,$$

while neither minimum is known.

The displayed universal-host quantifiers concern distance-one failure, not the classical exploration-trap parameter of Section 11.

Sharpness data for A^* . Minimal seven-cell traps of the extremal controller are not all trees; the exhaustive supplementary data also contain cyclic mechanisms. No detailed classification of those traps is needed for Theorem 10.3.

11 Related work

The relevant comparisons concern three different tasks: visiting every vertex or edge, meeting at one vertex, and the distance-one contact required here. They also involve different ways of supplying direction information and different orders of quantification over controllers, graphs, labellings and starts. We organize the comparison around these distinctions, rather than treating all finite-state walkers as one model.

Labyrinth automata, quantitative traps, and universal hosts. Budach’s work on labyrinth exploration [2] and the universal-trap paper of Antelmann, Budach and Rollik [1] are central historical predecessors. For their historical theorem statements we rely on the account of Kilibarda, Kudryavtsev and Ušćumlić [19], not on uninspected originals. Hemmerling’s monograph explicitly treats traps with a specified starting position and estimates their size in terms of internal states [13, pp. 133–134]. Thus neither state-dependent trap size nor a controller-dependent adversary is a new organizing principle.

The observation interface can also be quite close. Hoffmann’s inspected opening pages define plane mazes as full sublabyrinths of the square lattice, with free compass directions observed locally, and state a one-pebble exploration impossibility [14, pp. 433–434]. His finite-automaton definition uses a legal move at every step. Induced geometry and compulsory motion therefore do not, by themselves, separate our problem from that literature. What must still be transferred is the objective: an unvisited vertex is not a certificate that two moving copies always remain nonadjacent.

A quantitative predecessor makes this limitation especially concrete. The survey [19, Theorem 4, p. 25] attributes to Antelmann a proper planar exploration trap of size at most $C \exp \sqrt{2s \log_2(2s)}$ for an initialized s -state automaton. The cited statement does not specify a numerical C . Antelmann’s original article and dissertation remain uninspected; neither an exact low-state constant nor a rendezvous conclusion is inferred from this asymptotic bound.

Universal traps require an equally explicit qualification. Kilibarda proves universal exploration traps for finite families of automata [17, Theorems 1.2–1.3]. His later reduction paper gives a finite mosaic trap for each finite collective of independent automata and, in a separate theorem, one *infinite* mosaic trap for all such collectives [18, Theorems 3.5 and 3.8]. Consequently our finite obstruction menu and fixed host are not presented as the invention of universal trapping. The specific assembly preserves the complete masks queried by two executions, produces an explicitly bounded finite induced subcubic grid tree, and permits only the bad starts to depend on the controller. Its failure predicate is perpetual nonadjacency. The earlier exploration theorems are genuine quantifier predecessors, but none of the cited statements supplies that entire conclusion.

Finite-state exploration, exact small memory, and periodic traversal. Graph exploration supplies both asymptotic and exact-state comparisons. The work of Fraigniaud, Ilcinkas, Peer, Pelc and Peleg [9] studies port-based automata; a modern account of the space-complexity setting is given by Disser and Klimm [7]. The latter also makes explicit why vertices with the same observed incident colours are indistinguishable. This is a useful neighbor of our repeated-mask requirement, although exploration is a one-agent objective and its label conventions are not an absolute compass.

The closest numerical trap bound is particularly important to acknowledge. Ilcinkas [15, Theorem 2.1, p. 52] constructs, for every K -state port-based Moore automaton and $d \geq 3$, a simple-graph exploration trap with at most $K + d + 2$ vertices and maximum degree d . Substituting $K = 2, d = 3$ gives seven. The construction leaves a *vertex* unvisited, so an edge-versus-vertex slogan would not distinguish it from our theorem. Instead, a transfer must preserve the controller interface and induced-grid geometry and must create a nonadjacent pair of executions. No such state-preserving transfer is established here.

Exact one- and two-state classifications are also already available, rather than being a new kind of question. Fraigniaud, Ilcinkas and Pelc [10] classify regular graphs explorable by port-Mealy automata: the one-state class consists of the one-vertex graph, K_2 and cycles, while the two-state class consists of cliques and cycles. An automaton for a graph must work for every port assignment and initial position. These are strong predecessors for the small-memory viewpoint, but not determinations of our max–min trap parameters for a fixed compass-mask interface.

For the structural results, Diks, Fraigniaud, Kranakis and Pelc [6] already use a degree-based state lower bound and the cyclic next-port rule for perpetual tree exploration. Kilibarda’s Proposition 1.1 [17] states that every rotation system of a tree induces one contour; the same paper gives exact small-state exploration thresholds in a richer local-vision model. Thus the contour mechanism in Section 6 is classical. Our four-state table stores the incoming *compass* port internally, because it is not supplied as an observation; the radius-one rendezvous conclusion additionally uses path phase contact. This is not a per-vertex rotor-memory model: the fixed tree has no writable local state.

Periodic exploration is another close, but different, extremal problem. Ilcinkas [16] gives a three-state explorer with period at most $4n - 2$ after a suitable port assignment. That problem minimizes a guaranteed exploration period while permitting the designer to choose local orientations; its configurations include an incoming port. Our sharp-period theorem instead maximizes the least tagged period over allowed compass paths and controllers, under an internal-state budget. Neither the periodic-exploration bound nor the standard contour alone gives the weighted capacity and sharp all-state formula proved in Section 3.

Meeting identical agents: resources and port quantifiers. Pelc’s survey provides the broader deterministic network-rendezvous context [22]. A particularly relevant compass-labyrinth result is Kilibarda’s type-meeting theorem [20, Theorem 10]. Its strong version uses two copies of one collective and same-vertex contact; the minimal universal types for finite mosaic labyrinths are $(1, 2)$ and $(2, 0)$.

A type counts active automata and pebbles, not internal states. In particular, copying a collective of type $(2, 0)$ is not a theorem about two single agents. The negative side concerns same-vertex meeting and does not supply a distance-one trap or the exact values four, seven or eight in our model.

Bounded-memory rendezvous in port-labelled graphs is more developed than a comparison based only on a few constants would suggest. Czyzowicz, Kosowski and Pelc [3] establish log-space rendezvous bounds in arbitrary graphs. For trees, the distinction between the following two domains is essential. Fraigniaud and Pelc [11, Definitions 1.1–1.2 and Section 1.2] require success for every port labelling and restrict admissible initial pairs by perfect symmetrizability. Their simultaneous-start result does not cover every pair that happens to be nonsymmetric under one fixed labelling. Czyzowicz, Kosowski and Pelc [4] prove a logarithmic memory lower bound even for simultaneous starts on lines in that latter, broader domain. The inspected author versions explicitly expose the correction to the earlier conference formulation. These are not conflicting bounds once the domains are kept distinct.

Those models supply arrival-port information, permit waiting, and require same-node meeting. Our compass, compulsory motion, identical initial states and integer-time distance-one event must therefore all be retained when comparing the path threshold or the still-open three-state case. In particular, a lower bound on same-node rendezvous is not automatically a lower bound on distance-one contact, even if it already applies to simultaneous starts.

Recent nearby models. The 2026 preprint of Das and Pelc [5] is a close current finite-state comparison: it asks for a universal deterministic automaton on all feasible rendezvous instances of port-labelled graphs. Its negative result with finitely many stationary pebbles already has simultaneous-start line witnesses. Thus start delay alone cannot explain the separation from our positive path result. The same-node predicate, port observations and feasibility quantifiers remain decisive; the preprint does not settle our three-state question.

Recent grid and neighborhood work illustrates two other boundaries. Gao and Pelc [12] study identical agents with a coherent compass on the infinite grid, but allow waiting and permanent footprints, and require a common node. Eguchi, Kitamura and Izumi’s *neighborhood rendezvous* [8] assumes initially adjacent agents and still seeks same-node meeting; it does not mean success at distance one. Its randomized, labelled-graph setting is also not a fixed small-state anonymous controller. These papers clarify model choices rather than provide numerical predecessors to our trap thresholds.

Graph-walking realization and deterministic synthesis. Local descriptions of finite-state computations have strong precedents. Martynova [21] encodes graph-walking computations by attaching incoming and outgoing state sets to directions and imposing node consistency. Theorem 5 of the inspected author version transforms a *given* automaton into a signature describing accepted graphs. Here a graph and a cyclic scenario are prescribed, while the common transition table is unknown. This reverses the synthesis direction; it does not make the use of local state annotations itself new.

Similarly, Ulyantsev, Buzhinsky and Shalyto [23] identify deterministic machines from scenarios using state assignments and transition consistency. Our occurrence-colouring formulation belongs naturally to this exact-synthesis neighborhood. The content specific to Theorem 7.2 is a normal form for a restricted recurrent scenario: arbitrary row outputs are eliminated in favor of a finite common mask/difference profile and one bit per internal vertex. It is not a general theorem that scenario synthesis is new or that local annotations guarantee global realizability without further constraints.

Algebraic consistency and the scope of the contribution. Signed-graph balance and switching are classical; see Zaslavsky [24, Section 2, Proposition 2.1]. Gain graphs generalize edge signs to group elements, with identity product around balanced cycles [25]. This is the correct context for our signed-graph special cases, but not for the entire realization criterion. A profile system may have rows involving more than two bits; it balances all linear dependencies of one matrix. Successor differences may be zero, and the existential choice of one common profile can produce non-affine phase projections. Replacing the full criterion by one signed graph or by permutation-only transport would change the theorem.

Nor is polynomial recognition at a fixed state budget and alphabet asserted as a new complexity result: the set of complete controller tables is already finite independently of path length. The contribution claimed here is the proved combination of the exact model-specific trap thresholds, weighted recurrent-support bounds with sharp attainment, the special affine realization normal form, and observation-preserving obstruction assembly. The inspected predecessors do not supply this combination under the same hypotheses. That comparison is not an absolute priority claim; the access limitations on historical originals remain explicit.

12 Open problems

The remaining questions concern the interface between recurrence, accessibility and uniformity, rather than a missing numerical adjustment to the proved thresholds. The three-state path question is

$$F_{\mathcal{P},1}(3) ?$$

Equivalently, does one three-state controller solve every finite induced path, or does the eventual path state complexity become four? The realization theorem decides whether a prescribed contour-plus-matching cycle can exist. It does not decide whether one fixed controller has the required state-zero-accessible basin structure on every path. The unrestricted-grid quantity $F_{\mathcal{G},1}(3)$ is also open; branching invalidates the path phase-contact argument even on a unique recurrent cycle.

The exact minimum universal-host size u_2 and its subcubic-tree analogue $u_{2,\text{tree}3}$ are unknown. The current range $8 \leq u_2 \leq u_{2,\text{tree}3} \leq 5212$ combines a computer-assisted lower bound with an explicit construction. It neither proves that 5212 is optimal nor transfers the individual-controller threshold seven to the one-host quantifier order. Let b_2 be the minimum cardinality of an arbitrary finite complete test family. The certified selection gives $b_2 \leq 44$; the packing witness proves optimality only inside the fixed 144 candidates, and the global minimum remains open.

Beyond one state there is no exact higher-memory radius law here. The sharp period formula determines the maximum recurrent period over path geometries, not the realizability of every weighted walk by a common table. A full higher-state realization theory, including arbitrary four-state path normal forms, also remains open. These questions should preserve the distinctions between least tagged period and spatial period, between actual masks and support masks, and between a constructed cycle and its accessibility. None is resolved by extending a finite scan horizon.

A Local obstructions for the upper bound

This appendix supplies the mixed-axis and corner-bit arguments used in Section 5.1. Coordinates, masks, and next-state bits have the notation of that section. Every pair of starts below is in state zero. All displayed geometries are induced paths: for an L or U this follows from its defining coordinates, and for each listed endpoint order no nonconsecutive pair is at unit distance.

A.1 Excluding one reset axis

Lemma A.1. *A path survivor cannot have vertical straight map C_0 and horizontal straight map I in the frame of Lemma 5.2.*

Proof. Here $\delta(0, \text{NS}) = (\text{S}, 0)$, $\delta(1, \text{NS}) = (\text{N}, 0)$, and $n_0 = e_0 = 1$. We force three corner rows and then contradict survival. Every unlisted corner or endpoint bit remains arbitrary.

The ES state-zero row must be $(\text{S}, 0)$. If it chooses east, use $L(\text{S}, \text{E}; 3, 2)$ with starts S_2, C . The southern agent stays in the reset pocket S_1, S_2, S_3 . The corner-started agent stays in C, E_1, E_2 : every return to C from the horizontal straight cell has state zero, and that corner row sends it east. The disjoint regions and even-colored starts give a trap. If instead the row is $(\text{S}, 1)$, use $L(\text{S}, \text{E}; 4, 1)$ with starts S_4, C . The southern agent stays in S_2, S_3, S_4 , with S_2 visited only in state zero: from S_3 either state reaches S_2 or S_4 in state zero, and the endpoint returns to S_3 in either state. The other agent has exact cycle $C_0 \rightarrow (S_1)_1 \rightarrow C_0$. The regions are disjoint and the starts are even. Only $(\text{S}, 0)$ remains.

The SW state-zero row must point west. Use the upward U with endpoint order

$$(0, 0), (0, 1), (0, 2), (1, 2), (2, 2), (2, 1), (2, 0)$$

and starts $(0, 1), (2, 1)$. Every arrival at either upper corner from its straight vertical arm has state zero. The ES corner returns south by the row just forced. If the SW state-zero row also chose south, both vertical columns would be closed and two units apart. Thus SW in state zero chooses west.

The SW state-one row must be $(S, 0)$. If it chooses west, use $L(S, W; 3, 2)$ with starts S_3, W_1 . The pocket's closed tagged set contains $(S_3)_0$. The other agent begins

$$(W_1)_0 \rightarrow (W_2)_0 \rightarrow (W_1)_1 \rightarrow C_1,$$

using $e_0 = 1$. Every later arrival at C from the west arm also has state one, so that arm, including C , is closed. The regions are disjoint and both starts are odd. If the row is $(S, 1)$, use $L(S, W; 4, 2)$ with starts S_4, W_2 . The southern agent remains in S_2, S_3, S_4 as above. The other begins $(W_2)_0 \rightarrow (W_1)_1 \rightarrow C_1$ and then

$$C_1 \rightarrow (S_1)_1 \rightarrow C_0.$$

The already forced west-pointing state-zero row returns it to the west arm; every return from that arm reaches C_1 . Thus its whole trajectory lies in W_2, W_1, C, S_1 , with S_1 visited only in state one. These regions are disjoint and the starts are even. This excludes the second alternative and forces $(S, 0)$.

The forced rows give a trap. Use

$$(0, 0), (0, 1), (0, 2), (1, 2), (2, 2), (2, 1)$$

with starts $(0, 1), (2, 1)$. The left three-cell column is closed by the vertical reset and the ES state-zero row. The right agent has exact cycle

$$(2, 1)_0 \rightarrow (2, 2)_1 \rightarrow (2, 1)_0$$

by $n_0 = 1$ and the SW state-one row. Their horizontal coordinates remain zero and two, respectively. This contradicts path survival. $\square$

Lemma A.2. *A path survivor cannot have vertical straight map C_1 and horizontal straight map I in the frame of Lemma 5.2.*

Proof. Now $\delta(0, NS) = (S, 1)$, $\delta(1, NS) = (N, 1)$, and $e_0 = 1$. Reset pockets point north. We keep the initial state zero throughout; in particular, the free bit s_1 is treated in both of its cases.

The NW state-one row must be $(N, 1)$. If it chooses west, use $L(N, W; 3, 2)$ with starts N_2, W_2 . The north pocket is closed, while the other agent starts $(W_2)_0 \rightarrow (W_1)_1 \rightarrow C_1$. Every later arrival from the west arm has state one, so its west-pointing corner row closes W_2, W_1, C . The regions are disjoint and the starts are even. If the row is $(N, 0)$, use $L(N, W; 4, 1)$ with starts N_3, W_1 . The first agent stays in N_2, N_3, N_4 : the closed pocket tagged set is $\{(N_2)_1, (N_3)_0, (N_3)_1, (N_4)_1\}$. The second has prefix $(W_1)_0 \rightarrow C_1$ followed by $C_1 \rightarrow (N_1)_0 \rightarrow C_1$. The disjoint regions and odd starts again give a trap. This leaves only $(N, 1)$.

The NE state-one row must point east. Use the downward U, namely $U(N, E)$, with starts $(0, 1), (2, 1)$. Both arrivals at the lower corners from their vertical arms have state one. The right NW corner returns north. If the left NE state-one row also chose north, both columns would be closed and two units apart. It must therefore choose east.

The NE state-zero row must be $(N, 1)$. If it also chooses east, use $L(N, E; 3, 2)$ with starts N_2, E_2 . The north pocket is closed, and both corner states point east, closing C, E_1, E_2 . The regions are disjoint and the starts even. If the row is $(N, 0)$, use $L(N, E; 4, 2)$ with starts N_4, E_2 . The first endpoint move reaches N_3 in either state. From there, state zero reaches $(N_2)_1$ and state one reaches $(N_4)_1$; thereafter N_2, N_3, N_4 remains closed, with N_2 only in state one. The second agent stays in C, E_1, E_2, N_1 . Its only northern excursion is $C_0 \rightarrow (N_1)_0 \rightarrow C_1$, and the state-one corner row sends it east. The regions are disjoint and the original starts even. Hence the remaining row is $(N, 1)$.

Both values of s_1 now give a trap. For $s_1 = 0$, use

$$(0, 1), (0, 0), (1, 0), (2, 0), (2, 1), (2, 2)$$

with starts $(0,0), (2,2)$. The left agent has exact cycle $(0,0)_0 \rightarrow (0,1)_1 \rightarrow (0,0)_0$. The right column is closed because every corner arrival from its straight cell has state one and the NW row returns north. The columns are two units apart from time zero. For $s_1 = 1$, use $U(N, E)$ with starts $(0,0), (2,1)$. The complete entry prefixes are

$$\begin{aligned} (0,0)_0 &\rightarrow (0,1)_1 \rightarrow (0,2)_1, \\ (2,1)_0 &\rightarrow (2,0)_1 \rightarrow (2,1)_1 \rightarrow (2,2)_1. \end{aligned}$$

Each is then followed by its upper edge in state one. Again the agents remain in their separate columns throughout. These two cases exhaust the endpoint bit and contradict path survival. $\square$

A.2 The four corner output bits

Lemma A.3. *Assume identity straight maps, $n_0 = e_0 = 1$, and the direction conclusions*

$$\text{dir } \delta(0, \text{NE}) = \text{E}, \quad \text{dir } \delta(0, \text{ES}) = \text{S}, \quad \text{dir } \delta(1, \text{ES}) = \text{E}, \quad \text{dir } \delta(1, \text{SW}) = \text{S}.$$

If the controller survives all induced paths through seven cells, their output states are, respectively, 1, 0, 1, 0.

Proof. Each of the following six-cell paths excludes one wrong bit, independently of the other three output-bit conclusions.

ES in state zero cannot output one. Suppose $\delta(0, \text{ES}) = (\text{S}, 1)$ and use

$$(1, 2), (0, 2), (0, 1), (0, 0), (1, 0), (2, 0)$$

with starts $C = (0,0)$ and $U = (0,2)$. Write $M = (0,1)$, $H = (1,0)$, $E = (2,0)$, and $P = (1,2)$. The first agent stays in C, H, E : every return from H reaches C in state zero, whose NE row points east. The second has closed tagged set

$$\{U_0, U_1, M_1, P_0, P_1\}.$$

Indeed $U_0 \rightarrow M_1 \rightarrow U_1$, the ES state-one row points to P , and endpoint P returns to U in either state. The two spatial regions are disjoint and the starts even, proving failure.

ES in state one cannot output zero. Suppose $\delta(1, \text{ES}) = (\text{E}, 0)$ and use

$$(0, 1), (0, 2), (1, 2), (2, 2), (2, 1), (2, 0)$$

with starts $P = (0,1)$ and $I = (2,1)$. Put $C = (0,2)$, $H = (1,2)$, $D = (2,2)$, $B = (2,0)$. The left prefix is $P_0 \rightarrow C_1 \rightarrow H_0 \rightarrow C_0$. At C_0 the ES row points south to P , whose endpoint return may reach either state of C . Thus P, C, H is closed, with H only in state zero. The right arm B, I, D is closed: every arrival at D from I has state one and its SW row points south. The regions are disjoint and both starts odd.

NE in state zero cannot output zero. Suppose $\delta(0, \text{NE}) = (\text{E}, 0)$ and use

$$(0, 3), (0, 2), (1, 2), (2, 2), (2, 1), (2, 0)$$

with starts $(1,2), (2,1)$. The first agent has exact cycle $(1,2)_0 \leftrightarrow (0,2)_0$. The second stays in the right three-cell column by the state-one SW direction and the corner-arrival rule. The spatial regions are disjoint and both starts odd.

SW in state one cannot output one. Suppose $\delta(1, \text{SW}) = (\text{S}, 1)$ and use

$$(0, 2), (1, 2), (1, 1), (1, 0), (2, 0), (3, 0)$$

with starts $(0,2), (2,0)$. The upper agent's complete prefix and cycle are

$$(0,2)_0 \rightarrow (1,2)_1 \rightarrow (1,1)_1 \rightarrow (1,2)_1 \rightarrow \dots,$$

using $e_0 = 1$. The lower agent stays in the horizontal arm $(1,0), (2,0), (3,0)$: every arrival at its NE corner $(1,0)$ has state zero and is sent east. The regions are disjoint and both original starts even. This excludes the last wrong bit. $\square$

B Complete short-twig maps

This appendix completes the local calculation used in Lemma 5.26. Attach a path $p - u - v$, where u has exactly the two neighbours p, v and v is a leaf. Write de for the directions $p \rightarrow u$ and $u \rightarrow v$. The condition $e \neq -d$ ensures that the three vertices are distinct and leaves exactly twelve words. We stop upon returning to p , so its other neighbours and outgoing rule play no role in this table.

de	Masks at u, v	$u_0 \rightarrow$	$u_1 \rightarrow$	$v_0 \rightarrow$	$v_1 \rightarrow$	Return or internal cycle
NN	NS, S	p_1	p_1	u_1	u_1	All return as 1
NE	ES, W	p_1	v_1	u_1	u_0	All return as 1
NW	SW, E	v_0	p_1	u_1	u_1	All return as 1
EN	NW, S	v_1	p_0	u_1	u_1	All return as 0
EE	EW, W	p_0	v_1	u_1	u_0	All return as 0
ES	SW, N	p_0	v_1	u_0	u_1	u_0, v_0 return; (u_1, v_1)
SE	NE, W	p_0	v_1	u_1	u_0	All return as 0
SS	NS, N	v_1	v_1	u_0	u_1	All reach (u_1, v_1)
SW	NW, E	p_1	v_0	u_1	u_1	u_0 returns; (u_1, v_0)
WN	NE, S	v_0	p_1	u_1	u_1	All return as 1
WS	ES, N	v_1	p_1	u_0	u_1	All return as 1
WW	EW, E	v_0	p_1	u_1	u_1	All return as 1

Table 14: Complete stopped maps for two-cell twigs. Every arrow is a substitution in Table 3. Parentheses in the final column denote the unique internal tagged cycle in the specified row.

For the NE row, the longest return is $v_0 \rightarrow u_1 \rightarrow v_1 \rightarrow u_0 \rightarrow p_1$; all other starting tags give suffixes of this route. The EE and SE rows have the same four-step form, returning to p_0 . In each of the remaining nonabsorbing rows, the printed arrows give a return in at most four steps. For ES, the tags u_1, v_1 form the displayed cycle and $v_0 \rightarrow u_0 \rightarrow p_0$. For SS, $v_0 \rightarrow u_0 \rightarrow v_1$ joins (u_1, v_1) . For SW, $v_1 \rightarrow u_1$ joins (u_1, v_0) , while $u_0 \rightarrow p_1$. These routes verify every transient and every internal recurrent component.

For a one-cell twig in direction d , the leaf has mask $\{-d\}$. The endpoint rows therefore return state one for $d \in \{N, W\}$, preserve the arriving state for $d = S$, and complement it for $d = E$.

The permitted square ports

Use the absolute square coordinates $a = (0, 0), b = (1, 0), c = (1, 1), d = (0, 1)$. In family Q there are at most two exterior cells and no second square. For a two-cell twig at cN , a west continuation would meet the northwest corner, so the only words are NN, NE. The same grid-edge check gives EE, EN at cE , NN, NW at dN , and WW, WN at dW . All these rows return every tag. This excludes an attractor in a dormant twig even for a trajectory that starts there.

The four active ports have the complete behaviour in Table 15. The entry tag is obtained from the square corner table. The one-cell column records the return of that entry tag; an arbitrary tag on a one-cell twig also returns immediately, possibly in the other state. The all-tag assertions in the two-cell column include all four tags, not merely the tag entering from the core.

Port	Entry tag	One cell	Two-cell word	Complete effect
aS	u_1	Return a_1	SS or SW	The entering tag reaches the internal edge cycle. In the second word only u_0 returns to a_1 .
aW	u_0	Return a_1	WW or WS	Every tag returns to a_1 .
bE	u_1	Return b_0	EE or ES	The first word returns every tag to b_0 . The second captures the entry tag; u_0, v_0 return to b_0 .
bS	u_0	Return b_0	SE or SS	The first word returns every tag to b_0 . The second captures every tag on its outer state-one edge.

Table 15: Active square-port returns and absorbers. The one-cell return uses the listed entry tag. The two-cell options exclude continuations that make a second square.

In the aS case with word SW , the returnable tag does not become trapped within the stopped twig map: the square sends it $u_0 \rightarrow a_1 \rightarrow b_1 \rightarrow a_0 \rightarrow u_1$, after which it enters the absorber. Likewise the returnable tags of the ES twig at bE return to b_0 , travel through the square back to b_1 , and are then sent to u_1 . These uses of the square core are part of the proof of Lemma 5.27, rather than extra assertions about an isolated twig.

C Affine realization: gauges, elimination and witnesses

C.1 Coordinate choices and unused rows

The nonzero vectors of $H = \mathbb{F}_2^2$ are the three states. Every permutation of them extends uniquely to an element of $GL(2, 2) \cong S_3$. Applying this map simultaneously to all state constants, successor differences and occurrence expressions gives an isomorphic profile system. Swapping the two local paired roles replaces z_v by $z_v + 1$; changing an anchor replaces comparisons by their sums. Thus the construction does not depend on these presentational choices. An independent permutation at one mask, without transforming the successor links to other masks, is not a symmetry. Nor can different occurrences of one absolute mask be independently rotated by grid symmetries.

At a load-two vertex the unused state is $b(a_M) + a_M(z_v + 1)$. It is unused at that vertex, not necessarily at the other vertices of mask M . When both paired input states occur somewhere at M , κ_M is their uniquely determined successor difference. When only one occurs, it is an auxiliary description of the observed transitions; it does not prescribe an unobserved output. In particular, the unobserved row must not be assigned an affine extrapolation that equals the zero vector. The arbitrary legal completion in the sufficiency proof avoids this error. Endpoint comparisons cannot be dropped even when both endpoints have load one: identical endpoint masks and states refer to the same controller row.

C.2 Exact completion by elimination

Fix one profile and partition its variables into retained variables x and variables y to be completed. Write the system as

$$A_x x + A_y y = b.$$

For a fixed x there exists a completion precisely when

$$\lambda^T A_y = 0 \implies \lambda^T A_x x = \lambda^T b.$$

Indeed, this is exactly the condition that $b - A_x x$ belong to the column space of A_y , by Gaussian elimination. This is the correct completion statement for load-two roles: additional constraints on retained roles can remain after elimination. Before fixing a profile the set of completions is a union of these affine sets and need not itself be affine. An automatic completion assertion based only on saturated correspondences is therefore not used.

C.3 Signed graphs and incidence pseudoforests

In a fixed profile, first combine repeated terms in each scalar equation and retain constant equations. If every remaining equation has at most two variables, form a signed multigraph with a reference vertex $*$ of value zero. Represent $z_u = b$ by an edge $u*$ of gain b , and $z_u + z_v = b$ by an edge uv of gain b . Parallel edges are retained. A constant $0 = 1$ rejects the system.

Proposition C.1 (Graph balance). *After constant equations are checked, such a system is consistent if and only if the XOR of the gains around every cycle is zero. In particular, a forest is consistent, and a component with one cycle has exactly its one parity condition.*

Proof. A solution telescopes around each cycle. Conversely, choose a root in each component and propagate values along a spanning tree, fixing the root of the component containing $*$ to zero. Each non-tree edge is satisfied precisely when its fundamental cycle has gain zero. This also proves that the

solution space has dimension $c - 1$, where c is the number of components including the one containing $*$. $\square$

The incidence graph of a general scalar system is instead bipartite, with variable vertices and equation vertices. Its being a forest is *not* by itself a consistency criterion: the two equations $z = 0$ and $z = 1$ have an incidence tree. Use the following exact reductions. Check constant equations; substitute every unary fixation; discard degree-zero variables as free; and remove a degree-one variable together with its unique incident equation, saving that equation for back-substitution. Each reduction preserves existence of a solution. A finite incidence forest has a leaf, so these reductions decide it completely. If each incidence component has at most one cycle, the residual nonempty core has degree two at every vertex and has equations

$$z_1 + z_2 = c_1, \dots, z_{k-1} + z_k = c_{k-1}, z_k + z_1 = c_k.$$

It is consistent precisely when $\sum_i c_i = 0$. The constants here are the *updated* constants after substitution, not necessarily the original ones. Solving the core and back-substituting reconstructs a solution. All these tests must use one shared profile across the components.

For the general system an inclusion-minimal inconsistent set of rows is a minimal binary dependency whose coefficients cancel and whose right-hand sides sum to one. To see the minimality assertion, a proper coefficient dependency with right-hand side one would itself contradict minimality; one with right-hand side zero could be removed, leaving a smaller contradiction. These row circuits, not necessarily graph cycles, are the general algebraic obstructions. A failed profile is not a failed specification: every admissible profile must fail.

C.4 The saturated NAND witness

For the specification of Proposition 7.4, the occurrence word is

$$(0, 1, 0, 1, 2, 3, 2, 3, 4, 5, 6, 7, 6, 7, 6, 5, 4, 3, 2, 1).$$

With ordinary state names 0, 1, 2, the following assignments, in this occurrence order, realize the three allowed pairs of saturated correspondence bits:

$$\begin{aligned} 00 : & (0, 0, 2, 1, 1, 0, 2, 1, 1, 1, 0, 2, 1, 0, 2, 0, 2, 0, 2), \\ 01 : & (0, 0, 2, 1, 1, 0, 2, 1, 1, 0, 2, 1, 1, 0, 0, 2, 2, 2, 0, 2), \\ 10 : & (0, 0, 2, 1, 1, 2, 0, 1, 1, 1, 1, 2, 0, 1, 2, 2, 0, 0, 2, 2). \end{aligned}$$

The x -bit compares the W-pairs at occurrences (1, 19) and (5, 17); the y -bit compares the E-pairs at (4, 6) and (10, 12). Reading the successors verifies that each repeated (q, M) has the same direction and next state. States at each spatial vertex are distinct. These explicit assignments establish the positive cases without an enumeration argument; the main-text four-state contradiction excludes 11.

C.5 The six-vertex staircase

Take directions ENENE, doubled edge indices 1, 3, and phases 0, 1, respectively. Its word is

$$(0, 1, 2, 1, 2, 3, 4, 5, 4, 3, 4, 3, 2, 1).$$

All four internal vertices are saturated. Put $w = q_{13} = q_{11}$ for the singleton W-state of mask NW, and $e = q_4 = q_6$ for the singleton E-state of mask ES. Let x compare the N-pairs (q_1, q_3) and (q_5, q_9) , and let y compare the S-pairs (q_2, q_{12}) and (q_8, q_{10}) . The N-rows have successor pairs (q_2, e) and (e, q_{10}) . Identity correspondence would force $q_2 = e$, contradicting the two different direction roles at one mask. Hence $x = 1$, and the swapped correspondence forces $q_2 = q_{10}$, so $y = 1$. The S-rows have successor pairs (q_3, w) and (q_9, w) ; $y = 1$ would force $q_3 = w$, again contradicting distinct roles. Thus $y = 0$. Equivalently these constraints contain

$$x = 1, \quad x + y = 0, \quad y = 0,$$

whose sum is $0 = 1$. This proves impossibility directly. The separate assertion that no smaller direction-consistent specification is impossible is an exact finite check, not part of this argument.

D Boundary cases and consequences of realization

D.1 One-edge supports with their actual masks

Lemma D.1. *Let a support consist of one edge uv , let its directional multiplicity be k , and read the actual ambient masks M_u, M_v . An s -state simple tagged cycle exists exactly when $k \leq s$ if $M_u \neq M_v$, and exactly when $2k \leq s$ if $M_u = M_v$.*

Proof. Each endpoint needs k distinct states. If the masks are equal, the two sets must be disjoint because their outgoing directions are opposite. These are the necessary inequalities. If the masks differ, use states $0, \dots, k-1$ at both ends and prescribe

$$\delta(i, M_u) = (d, i), \quad \delta(i, M_v) = (-d, i + 1 \bmod k),$$

where d points from u to v . If the masks agree, use disjoint sets with $\delta(i, M) = (d, k + i)$ and $\delta(k + i, M) = (-d, i + 1 \bmod k)$. These rows produce the required cycle and can be completed legally. $\square$

Thus a doubled edge is possible with three states when it is the entire path, but not when both its endpoints have the same straight ambient mask. The multiplicity-three exception on an entire one-edge path is outside the contour-plus-matching input of Theorem 7.2.

D.2 Two-state contours and a strict memory separation

For a support of at least three vertices, the two-state capacity law forces every edge multiplicity to be one. This is a necessary normal form, not an automatic realization theorem. Fix an order of the two ports of every internal mask M , and let $h_M \in \mathbb{F}_2$ be the outgoing state for its first port; the other state is $h_M + 1$. Choose endpoint states, giving at most four endpoint profiles. At internal occurrences group equal (M, d) and equate their successor states. At endpoints group equal (M, q) and do the same. The resulting equations in the h_M are scalar XOR equations, including constants.

This criterion is exact. Necessity follows because both states are used at each internal vertex and its two outgoing directions differ. For sufficiency, the two assigned states there are distinct, and each repeated controller key has both its direction and successor fixed consistently by precisely the stated groups. Complete unused rows legally. Thus one of these at most four systems is consistent if and only if the ordinary contour is realizable with two states. There are at most six internal masks, so this particular recognition construction is linear in the word length for the fixed compass alphabet.

A useful strict separation is the family with directions

$$E^a N^b W^c, \quad a, c \geq 2, \quad b \geq 3.$$

Its ordinary contour requires exactly three states. For the two-state impossibility, let α be the N-state of the NW corner and β the S-state of the SW corner. The shared E-row of the straight horizontal mask forces $\alpha = \beta$, by comparing its transitions into the two corners. The straight vertical N-row compared with the upper corner forces its state to be $1 - \beta$; the vertical S-row compared with the lower corner forces its state to be $1 - \alpha$. These are then equal, contradicting the two directions at a straight vertical vertex. For sufficiency, the following used rows give one three-state controller for the whole family; blank rows may be completed legally:

Mask	State 0	State 1	State 2
EW	(E, 0)	(W, 1)	—
NS	—	(S, 1)	(N, 2)
NW	(N, 2)	(W, 1)	—
SW	(S, 1)	—	(W, 1)
E	—	(E, 0)	—

Start at the lower left endpoint in state 1. The horizontal directions use states 0 and 1, the vertical directions states 2 and 1, and the two corners use states $(0, 1)$ and $(2, 0)$. Directly following these rows gives the contour and distinct tags at each vertex. Minimality of its eight-vertex representative among all two-state contour failures is only a finite-check assertion.

D.3 Proper intervals retain ambient ports

Suppose the prescribed support is an interval I of an ambient induced path P , with at least three vertices. Masks must still be read in P . Use the internal profiles of Section 7 and choose injective endpoint states. When an endpoint mask M also occurs internally, require

$$X_i = a_M \iff d_i = u_M.$$

For a paired-side endpoint, its chosen state is $b(a_M) + a_M t_i$ for a determined constant $t_i \in \mathbb{F}_2$. Include these endpoint occurrences in the same U/B comparisons as internal occurrences. If a mask occurs only at endpoints, group by (M, X_i) , reject a group with differing outgoing directions, and otherwise equate successors in that group. The one-edge case is Lemma D.1.

These additions give the exact boundary-aware version of the affine criterion. For necessity, the real controller gives the same singleton/paired partition whenever that mask occurs internally; it therefore obeys the direction gate and all comparisons. For sufficiency the gate checks directions at shared endpoint/internal keys, U/B checks their successors, and the remaining endpoint groups check all boundary-only keys. This exhausts every repeated row, while all prescribed directions stay in I . Legal completion is performed using the ambient masks. The system still has linear size and a bounded profile family. Deleting the exterior ports instead would change the realization question.

For a fixed known controller, its internal direction partition, singleton state and paired successor difference are already fixed. Equate each actual successor to its known row output (affine on the two paired inputs) and impose the endpoint rows directly. At most 36 endpoint injections remain. This is an exact linear recognition test for a prescribed support and word in that fixed controller, not a solution of the basin problem over all paths.

D.4 Full-support cycles and their residual competitors

Proposition D.2. *Let a three-state controller have a recurrent cycle C using every vertex of a path P with $n \geq 3$. Every other recurrent cycle has a one-edge support and length two. If R_v denotes the states unused by C at v , these other cycles are exactly the pairs*

$$(u, q) \mapsto (v, r) \mapsto (u, q), \quad uv \in E(P), \quad q \in R_u, \quad r \in R_v.$$

If the vertices unsaturated by C form an independent set, C is the unique cycle and every tag enters it within $3n - |C|$ steps.

Proof. Distinct cycles of a deterministic map have disjoint tags. The full-support cycle uses at least two states at each internal vertex. A second cycle with at least two support edges would need two further states at an internal vertex of its support, which is also internal in P , an impossibility. A one-edge competitor has an endpoint internal in P , where at most one state remains; its multiplicity is therefore one. This proves the exact residual-row test. If the unsaturated vertices are independent, no support edge can have residual states at both ends, so no competitor exists. A trajectory outside C cannot repeat a tag before entering C , since repetition would give another cycle. There are $3n - |C|$ tags outside it, giving the stated entrance bound. $\square$

When all internal vertices are saturated the independent-set hypothesis holds. In the three-state sharp-period family of Section 3.2, this gives entrance within two steps for even n and three for odd n . Once both agents are on the unique cycle, path phase contact applies. This consequence concerns that one explicit family of paths, not all path geometries.

The accessibility limitation is concrete already on the four-vertex path with directions ENE. Consider the same cycle in two controllers,

$$C = (0_2, 1_1, 2_1, 3_1, 2_2, 1_2).$$

Their rows in states 1 and 2 agree; state 0 differs as follows:

Mask	State 1	State 2	State 0, good	State 0, bad
E	(E, 1)	(E, 1)	(E, 1)	(E, 0)
NW	(N, 1)	(W, 2)	(N, 1)	(W, 0)
ES	(E, 1)	(S, 2)	(E, 1)	(E, 0)
W	(W, 2)	(W, 2)	(W, 2)	(W, 0)

At every other nonempty mask choose its first port in a fixed compass order and next state 0, completing both tables. The good controller sends every state-zero start to C in one step; the other unused tags do so too. For the bad controller, $0_0 \leftrightarrow 1_0$ and $2_0 \leftrightarrow 3_0$ are accessible cycles, whereas C is not state-zero accessible. Starts $0_0, 2_0$ alternate between spatial pairs $(0, 2)$ and $(1, 3)$, always at distance two. Thus the same realized cycle can coexist with different basin behavior.

D.5 Why path support is essential

Path support cannot be replaced by arbitrary tree support, even if the one-agent functional graph has a unique recurrent tagged cycle. For an explicit example, take the seven-cell triod

$$c = (0, 0), \quad n = (0, 1), \quad n_2 = (0, 2), \quad s = (0, -1), \quad s_2 = (0, -2), \quad e = (1, 0), \quad e_2 = (2, 0),$$

and the following three-state rows; unused rows may be completed arbitrarily.

Mask	state 0	state 1	state 2
NES	(N, 0)	(S, 1)	(E, 0)
NS	(N, 2)	(S, 1)	(N, 2)
EW	(E, 0)	(W, 0)	(E, 0)
S	(S, 1)	(S, 1)	(S, 1)
N	(N, 0)	(N, 0)	(N, 0)
W	(W, 1)	(W, 1)	(W, 1)

These rows give the unique recurrent tagged cycle

$$c_0, n_0, (n_2)_2, n_1, c_1, s_1, (s_2)_1, s_0, c_2, e_0, (e_2)_0, e_1.$$

Every off-cycle tag enters it in one step. Nevertheless the state-zero starts n_0 and s_0 , which are six phases apart, remain at Manhattan distance exactly two at all twelve phases. Thus branching invalidates the one-dimensional sign argument and marks a real obstruction to extending the two-state method directly to three states.

E Proof status and computational provenance

A deductive proof below does not depend on bounded search; it is not a claim of proof-assistant verification. A computer-assisted proof requires a finite certificate. Computations accompanying deductive theorems validate examples or implementations, not the all-size arguments.

Results	Proof basis	Computational dependency
Radius law; $F(1) = 4$; $F(2) = 7$ and grid-tree value; capacity, period and phase contact	Deductive	None; table replay is validation.
Four-state path universality; duality; fixed-horizon finiteness; affine realization and phase locking	Deductive	None; finite criterion/oracle agreement validates implementations.
Path value $F_{P,1}(2) = 8$	Computer-assisted	Analytic upper bound; exhaustive path-survival lower certificate.
Completeness of $\mathcal{U}_{\text{hum}}$	Deductive	144/76/22/13 are exact inventory counts.
Selected 44 tests; optimum within 144	Computer-assisted	Physical encoding and LRAT upper certificate; disjoint packing lower certificate.
Protected attachment, stretching and host assembly; exact-size extension	Deductive construction	Finite validation of 172 pieces, the 5,212-cell base and extension site.
$u_2 \geq 8$	Computer-assisted	Exhaustive small-maze check against eight fixed conjugates.

Exact finite scopes. The path lower certificate checks all 4,042 initially far unordered pairs on 337 translated shapes through seven cells, preserving absolute orientation (Section 6); it does not prove unrestricted path universality. The independent affine comparison covers 203,260 specifications through eight vertices: 94,852 realizable, 105,128 direction rejections and 3,280 residual rejections, with no criterion/oracle mismatch. Identifying path reversal halves this same corpus, not an independent sample. Saved full assignments, profiles and periods yield 189,704 reconstructed tables from two witness sets and checks of all 94,852 positive profiles. There are 243 normalized staircase certificates. The saturated NAND check has 20 normalized solutions, with counts 12, 4, 4, 0 for 00, 01, 10, 11; the displayed witnesses and contradiction, rather than these counts, prove the relation.

Certificate and geometry boundaries. The 144-test menu comes from terminal lemmas, not controller-space exhaustion. For the selected 44, LRAT verification must be accompanied by validation of the physical-instance encoding; packing proves a lower bound only within the candidates. The host has 5,212 vertices, 5,211 induced edges and maximum degree three. These finite facts validate its construction, not optimality.

Available replay. The supplied checks replay path survival, all 144 physical decision functions, semantic CNF reconstruction and the selected 44 LRAT certificate, the independent packing lower bound, and the universal lower-bound enumeration from A^* . They also validate all 172 ported fragments, host geometry and exact-size extension, and regenerate the finite realization corpus, complete controller witnesses and staircase certificates. All dependencies are included; no historical archive or network access is required. Finite validation does not replace any all-size deductive proof.

F Data availability and reproducibility

The source distribution includes complete LaTeX and TikZ sources, the 25-item bibliography and one canonical PDF. Supplementary data are organized as `paths/`, `obstructions/` and `realization/`. The release README specifies source-build and full-verification commands; manifests record file hashes, provenance and the claim-to-checker map. The default verifier runs every declared finite chain, regenerates the realization corpus, compares deterministic outputs and confirms that frozen inputs are unchanged. All generated files go to an explicitly selected external directory. A clean offline replay without historical directories passed all stages; the full success status is `PASS_FULL_REPRODUCIBILITY`. These checks preserve the proof-status boundaries in Appendix E.

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